Transformer Language Models

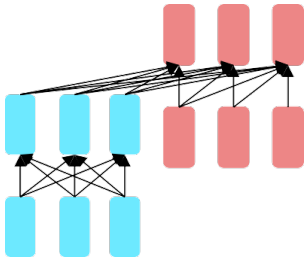
CSCI 601-771 (NLP: Self-Supervised Models)

https://self-supervised.cs.jhu.edu/fa2026/

After Transformer ...

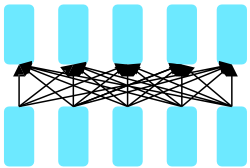


Topology variations



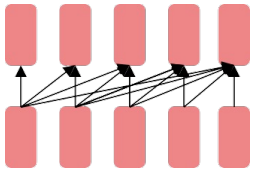
Encoder-Decoders

Encodes an input sequence and generates an output sequence autoregressively, with the decoder also attending to the encoder's representations.



Encoders

Builds contextual representations of input tokens by allowing each token to attend to all input positions.



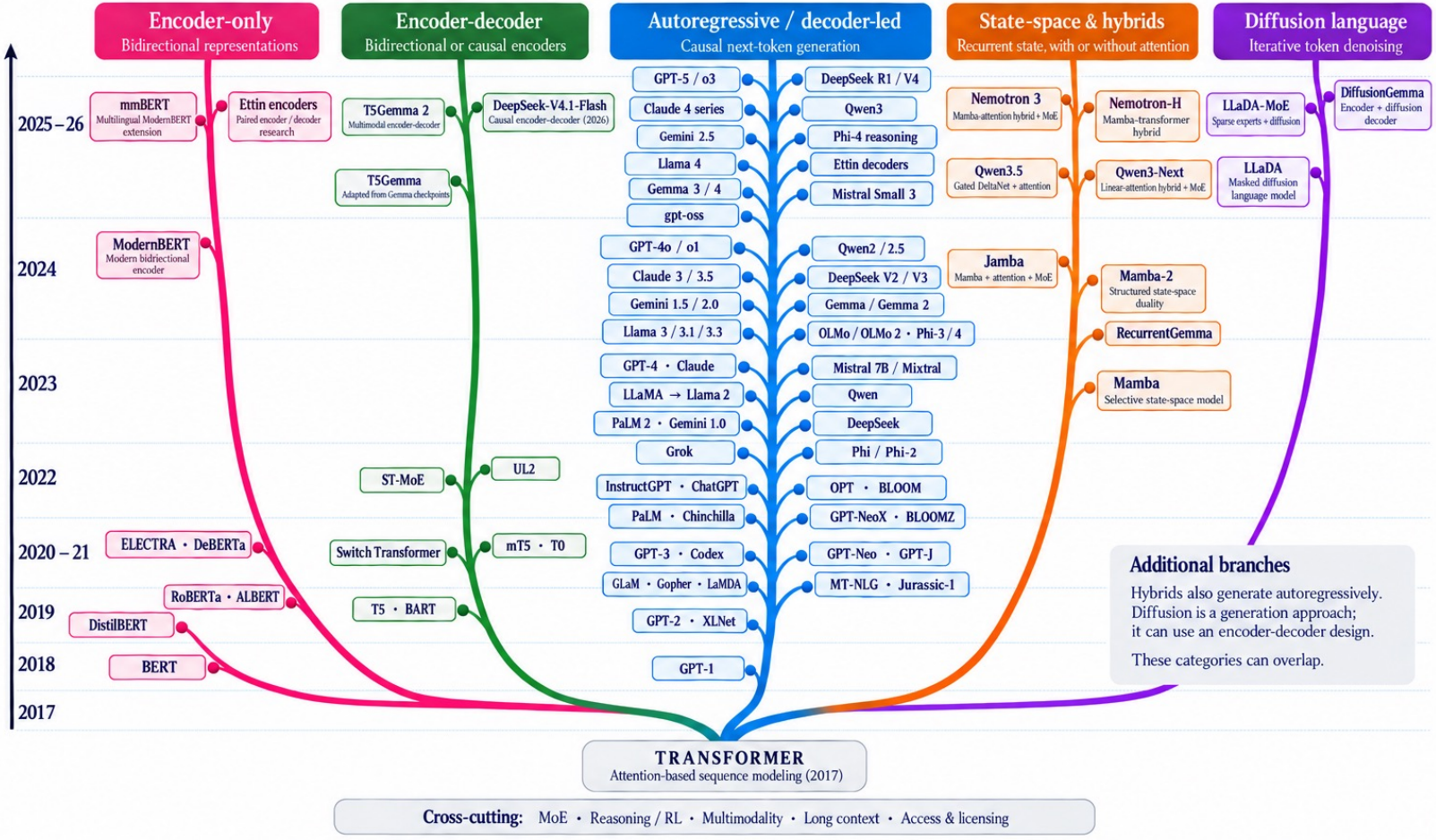
Decoders

Generates tokens autoregressively, with each position attending only to itself and earlier positions.

The Evolution of Language Models

Selected milestones • 2017 – 2026

Updated October 2026



Branches group approaches, not checkpoint ancestry. Categories can overlap; proprietary architectures may be undisclosed.

Inspired by Yang et al. (2023).

Other architectural variations

- **Topology:** Decoder-only, Encoder-decoder, Encoder-only.
- **Sequence mixing in each layer:**
  - Example: MHA, MQA, GQA, etc.
  - Patterns: global, sliding-window, or interleaved local/global.
- **Positional encodings:** e.g., RoPE.
- **Feedforward layers:** Dense vs. MoE, Activations: GELU, SwiGLU, etc.
- **Normalization and block layout:** pre-norm vs. post-norm.
- **Dimensions:** Depth vs. width, head count and head dimension.
- **Tokenization:** Vocab size and algorithm (e.g., BPE), tokenizer-free models
- **Output scheme:** Next-token prediction, multi-token prediction, Diffusion LMs

Legend: Already covered them; will cover next; will hopefully cover at some point.

Normalization

Quiz: Pre-norm vs Post-norm

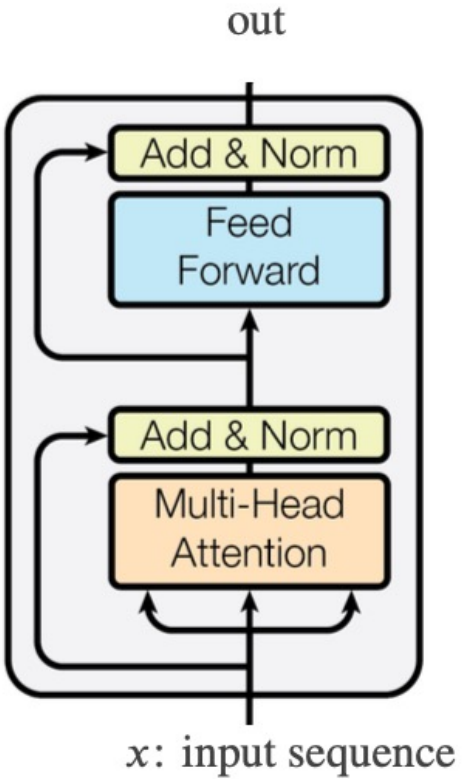
- Which is the original implementation?
- Which one gives gradients a direct identity path?

Post-norm

$$\text{LayerNorm}\big(x + \text{SubLayer}(x)\big)$$

Pre-norm

$$x + \text{SubLayer}\big(\text{LayerNorm}(x)\big)$$



Pre-norm vs Post-norm

- Pre-norm (right) is set up so that LayerNorm does not disrupt the residual stream (in gray)
- Both are valid architectures, but their optimization behavior differs.
- In practice, however, **Pre-norm is preferred over Post-norm**.

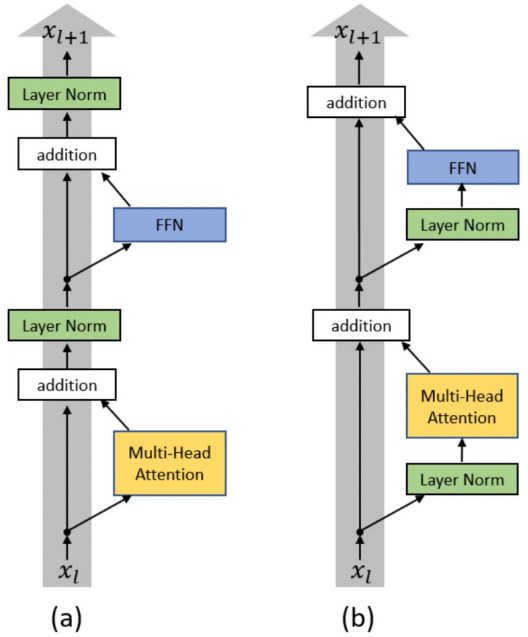
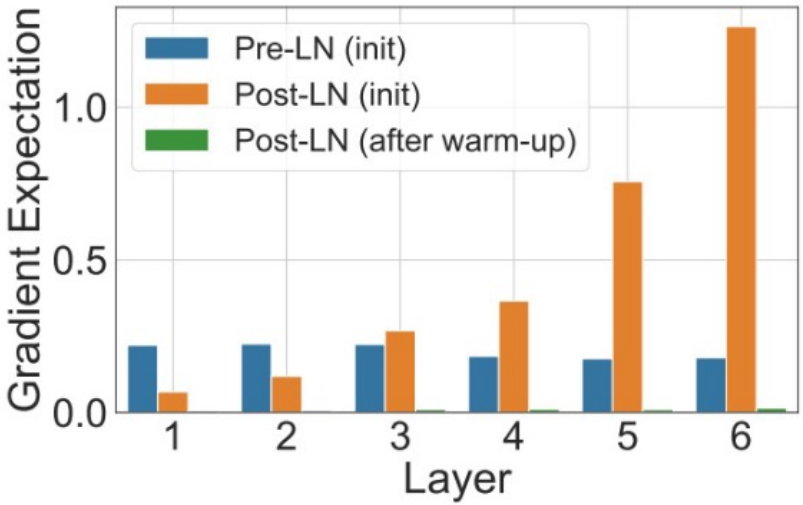


Figure 1. (a) Post-LN Transformer layer; (b) Pre-LN Transformer layer.

On Layer Normalization in the Transformer Architecture, 2020

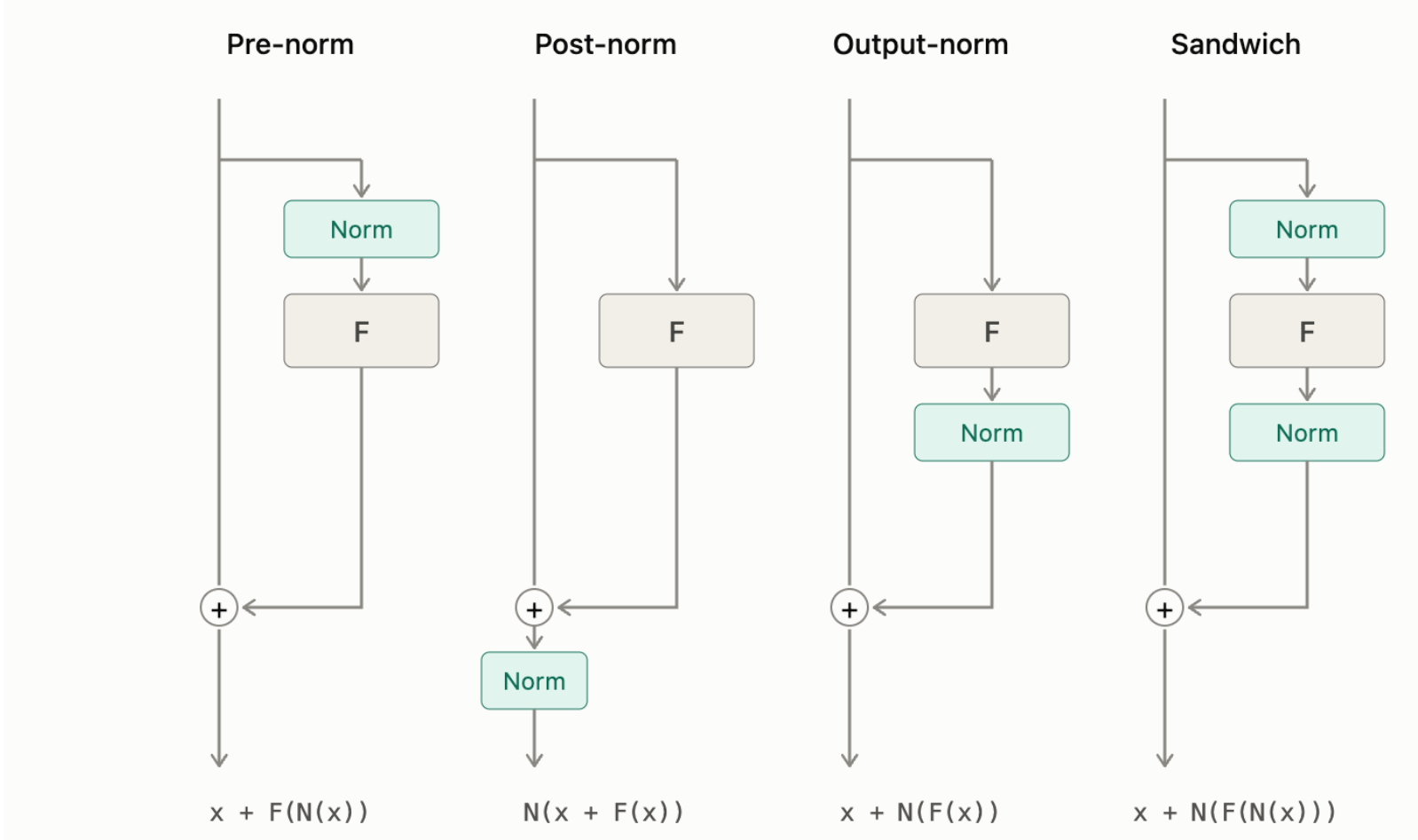
Pre-norm vs Post-norm: Empiric

More balanced gradients: Post-norm has large gradients near the output at initialization; pre-norm keeps them more uniform across layers.

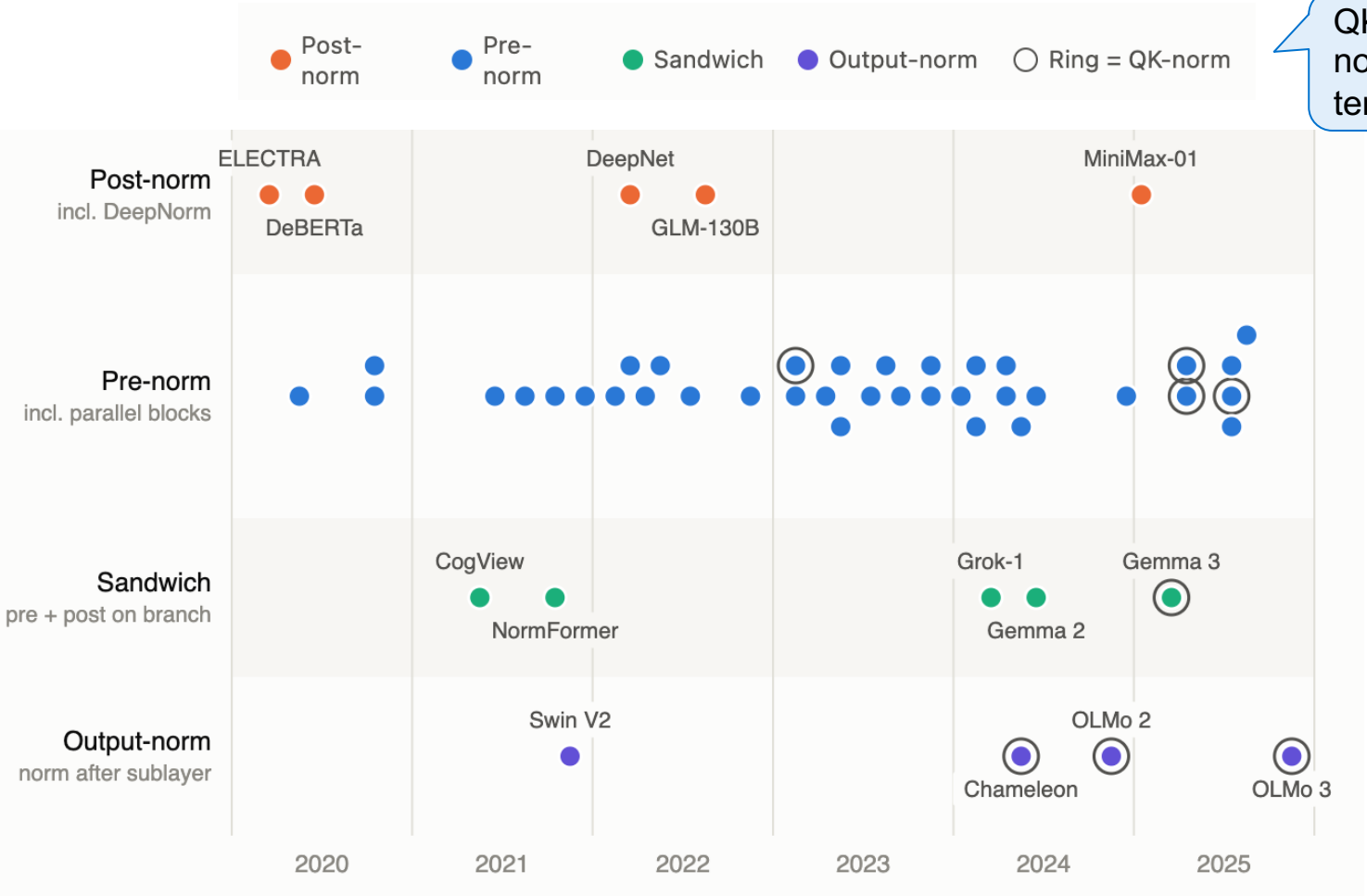


On Layer Normalization in the Transformer Architecture, 2020

Other forms of LayerNorm



Who uses what?



QK-norm also normalizes q and k tensors.

LayerNorm vs. RMSNorm

- Original transformer: **LayerNorm**
  - LayerNorm both centers activations and controls their scale.
  - What if we keep scale control **but skip centering**?

Stats computed across **each hidden dimensions**

$$\text{LayerNorm}(x) = \gamma \odot \frac{x - \mu(x)}{\sqrt{\text{Var}(x) + \epsilon}} + \beta.$$

- **RMSNorm:** Does not subtract mean or add a bias term

$$\text{RMSNorm}(x) = \gamma \odot \frac{x}{\sqrt{\frac{1}{d}\sum_i x_i^2 + \epsilon}}.$$

Why RMSNorm?

- **Modern explanation** – it's faster (and just as good).
  - **Fewer operations** (no mean calculation)
  - **Fewer parameters** (no bias term to store)

- Does this explanation make sense?
  - Matrix multiplies are the vast majority of FLOPs
  - Non-matmul ops only make up 0.2% of our FLOPS
    - So perhaps it doesn't matter that GPUs compute non-matmul ops slower.

"Tensor Contraction" := matmuls

Table 1. Proportions for operator classes in PyTorch.

Operator class	% flop	% Runtime
△ Tensor contraction	99.80	61.0
□ Stat. normalization	0.17	25.5
○ Element-wise	0.03	13.5

[Slide credit: Tatsu Hashimoto]

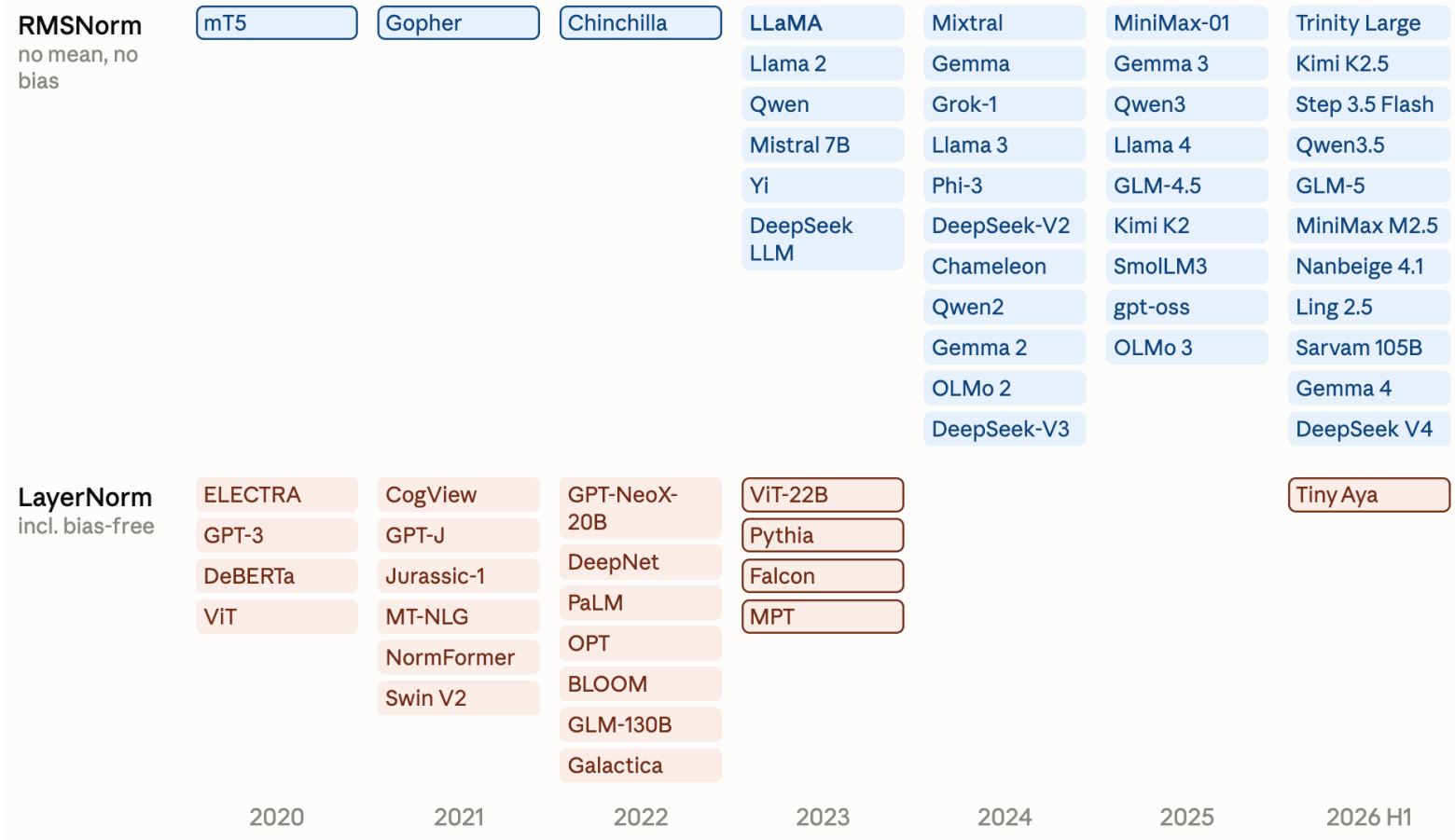
Why RMSNorm?

- **RMSNorm** runtime (and surprisingly, perf) gains have been seen in papers

Model	Params	Ops	Step/s	Early loss	Final loss	SGLUE	XSum	WebQ	WMT EnDe
Vanilla Transformer	$223M$	$11.1T$	3.50	2.182 ± 0.005	1.838	71.66	17.78	23.02	26.62
RMS Norm	$223M$	$11.1T$	3.68	2.167 ± 0.008	**1.821**	**75.45**	**17.94**	**24.07**	**27.14**

[Slide credit: Tatsu Hashimoto]

Who uses what?



LayerNorms: Recap

- Basically, everyone does pre-norm.
  - Intuition – keep the good parts of residual connections
  - Observations – nicer gradient propagation, fewer spike

- Most people do RMSnorm
  - In practice, works as well as LayerNorm
  - But, has fewer parameters to move around, which saves on wallclock time

[Slide credit: Tatsu Hashimoto]

FFN layers

$$\mathrm{FFN}(\mathbf{x}) = f(\mathbf{x}\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + \boldsymbol{b}_2$$

Is the Bias Term Necessary?

- FFN: The same nonlinear transformation independently to **each** token
- Most modern transformers don't have bias terms.
  - Original Transformer:

$$\text{FFN}(\mathbf{x}) = f(\mathbf{x}\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + \boldsymbol{b}_2$$

$$\boldsymbol{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}},$$
$$\boldsymbol{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$$

  - Most implementations (if they're not gated):

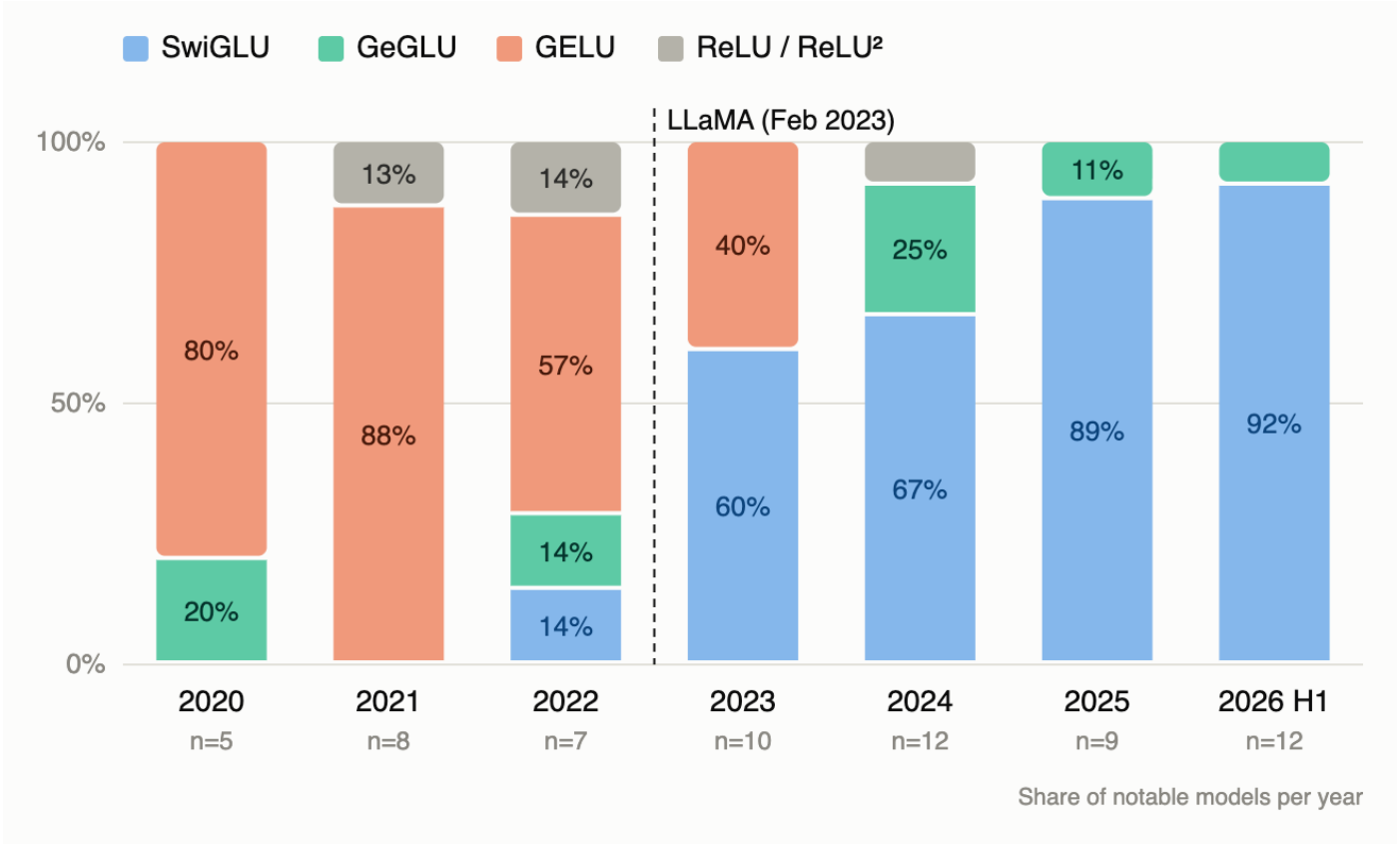
$$\text{FFN}(\mathbf{x}) = f(\mathbf{x}\boldsymbol{W}_1)\boldsymbol{W}_2$$

- Reasons: less memory (similar to RMSnorm).

[Slide credit: Tatsu Hashimoto]

Which Activation fun should we use?

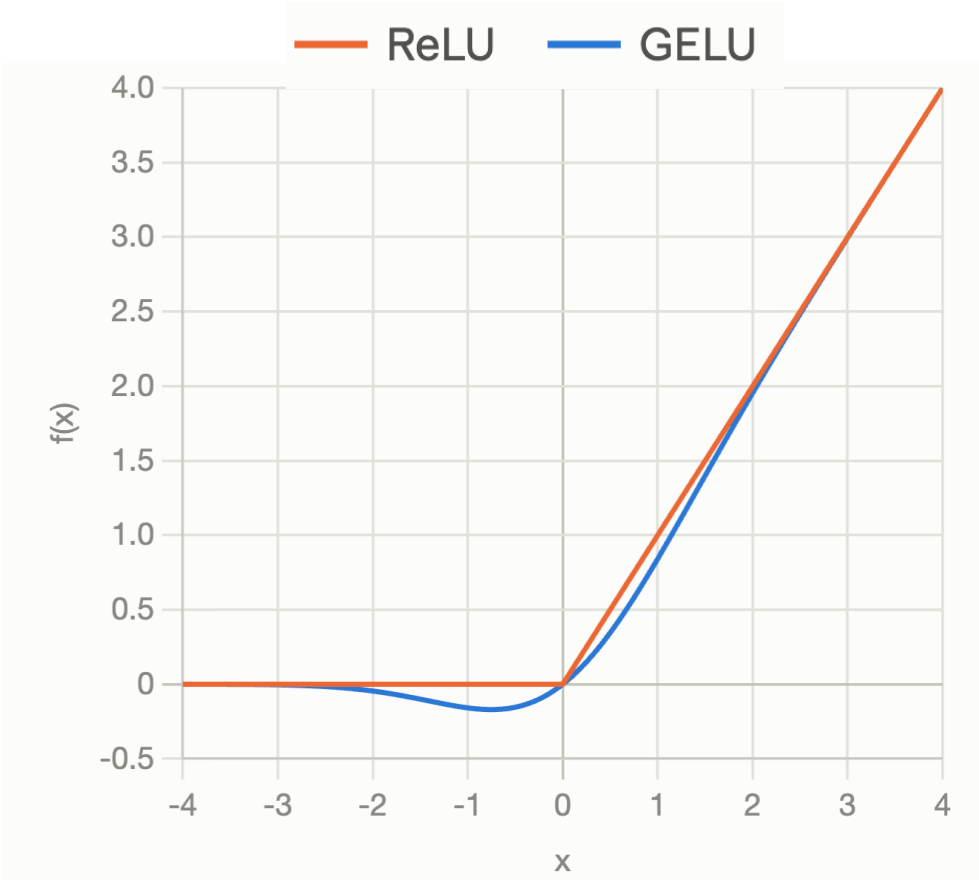
- There are many choices; but trending toward “gated” functions.



Activations: ReLU vs GELU

- **ReLU:**
  - **Notable models:** Original transformer, T5, Chinchilla

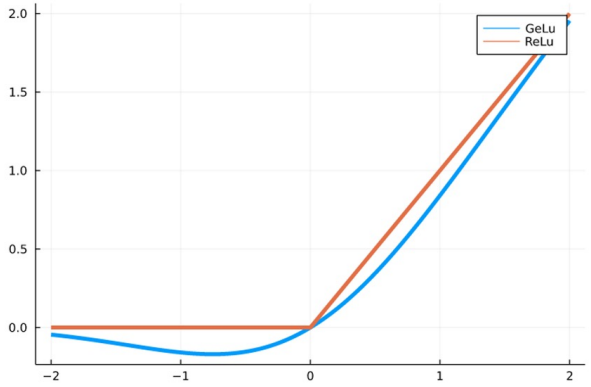
- **GeLU:**
  - **Notable models:** GPT1/2/3, BERT, RoBERTa, GPT2



Bonus

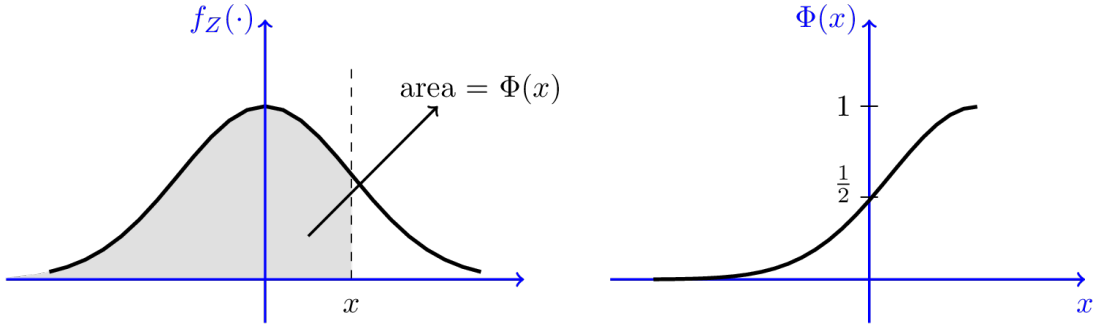
GELU function, in details

$$\mathrm{FFN}(\mathbf{x}) = \mathrm{GELU}(\mathbf{x}\boldsymbol{W}_1)\boldsymbol{W}_2$$
$$\mathrm{GELU}(\mathbf{y}) \coloneqq \mathbf{y}\Phi(\mathbf{y})$$



- Here $\Phi(\mathbf{y})$ the cumulative distribution function (CDF) of a normal distribution:

$$\Phi(\mathrm{y}) = \frac{1}{2}\left(1 + \mathrm{erf}\left(\frac{x}{\sqrt{2}}\right)\right)$$



Gated activations (*GLU)

- Gated activations modify the first part of the FFN layer:

$$\mathrm{FFN}(\mathbf{x}) = f(\mathbf{x}\boldsymbol{W}_1)\boldsymbol{W}_2$$

- What if the first layer depended on two different views of the input?

$$f(\mathbf{x}\boldsymbol{W}_1) \rightarrow f(\mathbf{x}\boldsymbol{W}_1) \odot (\mathbf{x}\mathbf{V})$$

  - The input is projected **twice**.
  - One projection, through the activation, acts as a **gate** (a soft volume knob), and it multiplies the other projection **elementwise**.

- The fully gated FFN is written in this form:

$$\mathrm{FFN}(\mathbf{x}) = (f(\mathbf{x}\boldsymbol{W}_1) \odot (\mathbf{x}\mathbf{V}))\boldsymbol{W}_2.$$

Language Modeling with Gated Convolutional Networks, 2017
GLU Variants Improve Transformer, 2020

Activations: Gated activations variants

- **GeGLU**

$$\mathrm{FFN}_{GeGLU}(\mathbf{x}; \boldsymbol{W}_1, \boldsymbol{W}_2, \boldsymbol{V}) = (\mathrm{GELU}(\mathbf{x}\boldsymbol{W}_1) \odot (\mathbf{x}\mathbf{V}))\boldsymbol{W}_2.$$

  - **Notable models:** T5 family, Gemma family (1 - 4)

- **SwiGLU:** swish function is $x \cdot \mathrm{sigmoid}(x)$:

$$\mathrm{FFN}_{SwiGLU}(\mathbf{x}; \boldsymbol{W}_1, \boldsymbol{W}_2, \boldsymbol{V}) = (\mathrm{Swish}(\mathbf{x}\boldsymbol{W}_1) \odot (\mathbf{x}\mathbf{V}))\boldsymbol{W}_2.$$

  - **Notable models:** LLaMA and nearly every LLM since 2023

Do Gated Linear Units work?

- Yes, fairly consistently so.

	Score Average	CoLA MCC	SST-2 Acc
FFN_{ReLU}	83.80	51.32	94.04
FFN_{GELU}	83.86	53.48	94.04
$\text{FFN}_{\text{Swish}}$	83.60	49.79	93.69
FFN_{GLU}	84.20	49.16	94.27
$\text{FFN}_{\text{GEGLU}}$	84.12	53.65	93.92
$\text{FFN}_{\text{Bilinear}}$	83.79	51.02	**94.38**
$\text{FFN}_{\text{SwiGLU}}$	84.36	51.59	93.92
$\text{FFN}_{\text{ReGLU}}$	**84.67**	**56.16**	**94.38**
[Raffel et al., 2019]	83.28	53.84	92.68
ibid. stddev.	0.235	1.111	0.569

GLU Variants Improve Transformer, 2020

Do gated linear units work?

- Yes, fairly consistently so.

Model	Params	Ops	Step/s	Early loss	Final loss	SGLUE	XSum	WebQ
Vanilla Transformer	$223M$	$11.1T$	3.50	2.182 ± 0.005	1.838	71.66	17.78	23.02
GeLU	$223M$	$11.1T$	3.58	2.179 ± 0.003	1.838	**75.79**	**17.86**	**25.13**
Swish	$223M$	$11.1T$	3.62	2.186 ± 0.003	1.847	**73.77**	17.74	**24.34**
ELU	$223M$	$11.1T$	3.56	2.270 ± 0.007	1.932	67.83	16.73	23.02
GLU	$223M$	$11.1T$	3.59	2.174 ± 0.003	**1.814**	**74.20**	**17.42**	24.34
GeGLU	$223M$	$11.1T$	3.55	2.130 ± 0.006	**1.792**	**75.96**	**18.27**	**24.87**
ReGLU	$223M$	$11.1T$	3.57	2.145 ± 0.004	**1.803**	**76.17**	**18.36**	**24.87**
SeLU	$223M$	$11.1T$	3.55	2.315 ± 0.004	1.948	68.76	16.76	22.75
SwiGLU	$223M$	$11.1T$	3.53	2.127 ± 0.003	**1.789**	**76.00**	**18.20**	**24.34**
LiGLU	$223M$	$11.1T$	3.59	2.149 ± 0.005	**1.798**	**75.34**	**17.97**	**24.34**
Sigmoid	$223M$	$11.1T$	3.63	2.291 ± 0.019	1.867	**74.31**	17.51	23.02
Softplus	$223M$	$11.1T$	3.47	2.207 ± 0.011	1.850	**72.45**	17.65	**24.34**

Recap: FFN layers

- Many variations (ReLU, GELU, *GLU) across models.

- **Gated activations (*GLU):** two projections; one projection gates the other elementwise.
  - **Gains:** Modest but consistent gains at equal compute (Shazeer, 2020).
  - **Cost:** three matrices instead of two, so d_ff shrinks to about ⅔ (4d → $\frac{8}{3}$d) to keep parameters equal. (more on this later)
  - Default in nearly all LLMs since 2023; SwiGLU is the most common.

- **Bias term:** People more generally drop bias terms since the compute/param tradeoffs are not great.

Architecture layout

Serial vs Parallel Layer

- The standard “serial” formulation: they compute attention, then the MLP
  - h = x + Attention(LayerNorm(x))
  - y = h + MLP(LayerNorm(h))

- The **parallel** formulation:
  - y = h + MLP(LayerNorm(x)) + Attention(LayerNorm(x))

  - Note about efficiency:
    - Both branches read the same LN(x), so one LayerNorm suffices.
    - The attention QKV and MLP up-projection can be fused into one matmul.

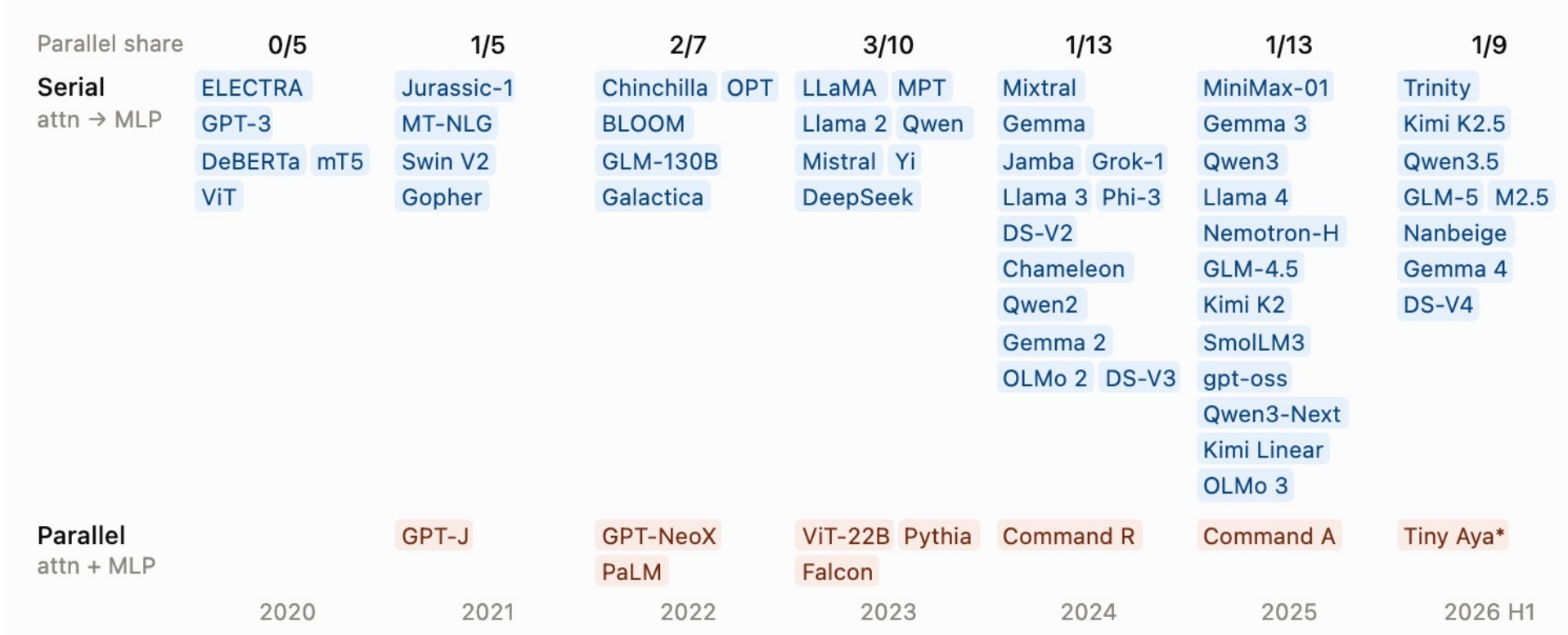
Proposed by GPT-J (Wang & Komatsuzaki, 2021)

Parallel Layer: do they work?

- The evidence is sparse.
- From PaLM paper: *"The parallel formulation results in roughly 15% faster training speed at large scales … Ablation experiments showed a small quality degradation at 8B scale but no quality degradation at 62B scale"*
  - the paper says it never ablated at 540B; it extrapolated from the 8B and 62B results that parallel layers should be quality-neutral at that scale.

PaLM: Scaling Language Modeling with Pathways, 2022

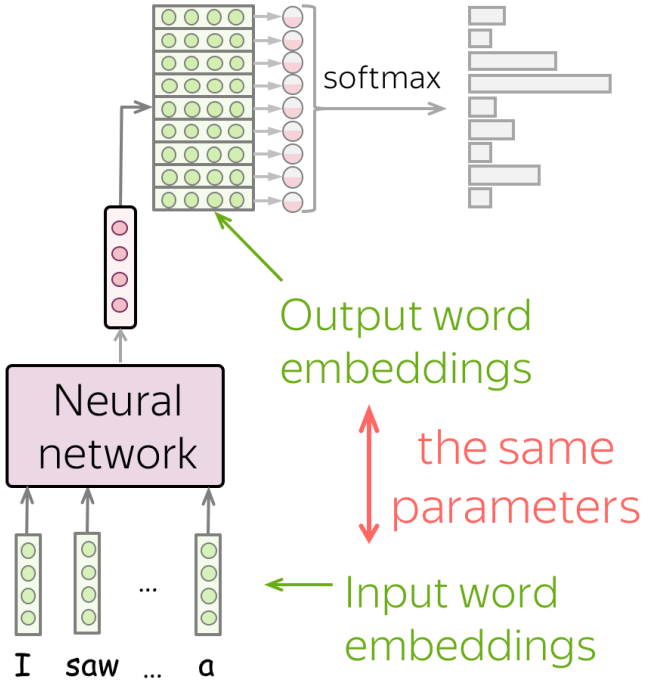
Serial vs Parallel Layer: who uses what?



Embedding parameter tying

- Embed matrix (vocab × d): turns token IDs into vectors
- Output projection (d × vocab): turns hidden states back into logits over the vocabulary.

- **Problem:** Both are vocab-sized and often among the largest single tensors in the model.
  - Their dimensions match (after a transpose).
  - Do we need to keep separate matrices?



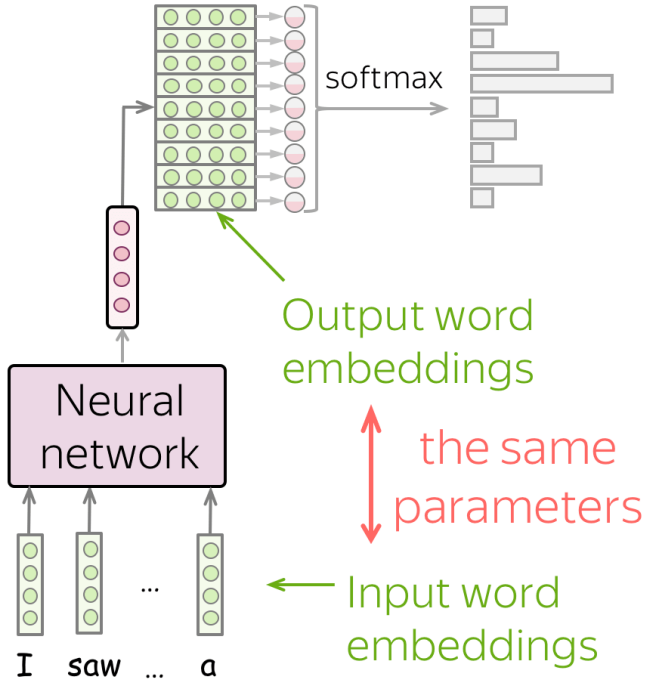
Using the Output Embedding to Improve Language Models, 2017

Embedding parameter tying

- The same weight matrix can be used for both the input embeddings and the output (projection) layer.

```
class TransformerWithTiedEmbeddings(nn.Module):
    def __init__(self, vocab_size, d_model):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, d_model)
        self.transformer = nn.Transformer(d_model=d_model)
        self.output_layer = nn.Linear(d_model, vocab_size)

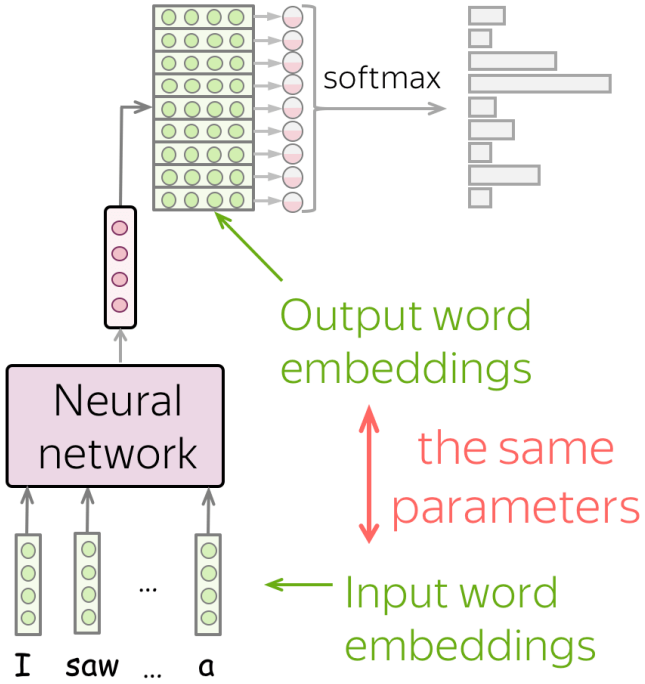
        # Tying embeddings
        self.output_layer.weight = self.embedding.weight
```



Using the Output Embedding to Improve Language Models, 2017

Embedding parameter tying: justification

- **Intuition:** Input and output embeddings should live in the same space. A token's meaning shouldn't depend on whether it's being *read* in or *predicted*.
- **Better embeddings for rare words:** The output matrix gets a gradient for *every* vocab token at every softmax step. The input embedding only updates for tokens in the batch. Tying lets rare words learn from that dense signal.
- **Memory efficiency:** Cuts one vocab × d matrix.
  - GPT-2 small: ~38M of 124M params (~30%).
  - Though the gains diminish for larger models.



Using the Output Embedding to Improve Language Models, 2017

Embedding parameter tying: who uses it?

Tied share	3/4	1/2	4/5	2/9	4/12	3/10	2/6
Tied shared in/out	ELECTRA GPT-3 DeBERTa	MT-NLG*	PaLM OPT BLOOM Galactica	Falcon* MPT	Gemma Command R Gemma 2 Llama 3.2	Gemma 3 Command A* SmolLM3	Tiny Aya* Gemma 4*
Untied separate head	mT5	GPT-J	GPT-NeoX	LLaMA Pythia Llama 2 Qwen Mistral Yi DeepSeek	Mixtral Nemotron-4 Llama 3 Phi-3 DS-V2 Chameleon* OLMo 2 DS-V3	MiniMax-01* Llama 4 GLM-4.5 Kimi K2 gpt-oss Qwen3-Next* OLMo 3*	Kimi K2.5 GLM-5* M2.5* DS-V4*
	2020	2021	2022	2023	2024	2025	2026 H1

Hover for details. * = inferred.

Recap so far

- **Serial vs parallel layers:**
  - Parallel layers have a compute win.
  - Empirical evidence is sparse.

- **Parameter tying:**
  - Reduces model parameters and leads to (modest) gains.
  - Unclear how much it matters at scale; many large LLMs (e.g., LLaMA) don’t tie.

Architecture Hyperparams

There are a ton of question regarding architecture hyperparameters:

- How much bigger should the feedforward size be compared to hidden size?
- How many heads? Should # of heads always divide hidden size?
- Should we make our model wide or deep?

The Surprising Consensus #1: FFN Dimension Ratio

- **Feedforward – model dimension ratio:**

$$\mathrm{FFN}(\mathbf{x}) = f(\mathbf{x}\boldsymbol{W}_1 + b_1)\boldsymbol{W}_2 + \boldsymbol{b}_2$$
$$\boldsymbol{W}_1 \in \mathbb{R}^{d \times d_{\mathrm{ff}}},$$
$$\boldsymbol{W}_2 \in \mathbb{R}^{d_{\mathrm{ff}} \times d}$$

- There are two dimensions that are relevant – the feedforward dim (d_{ff}) and model dim (d). What should their relationship be?

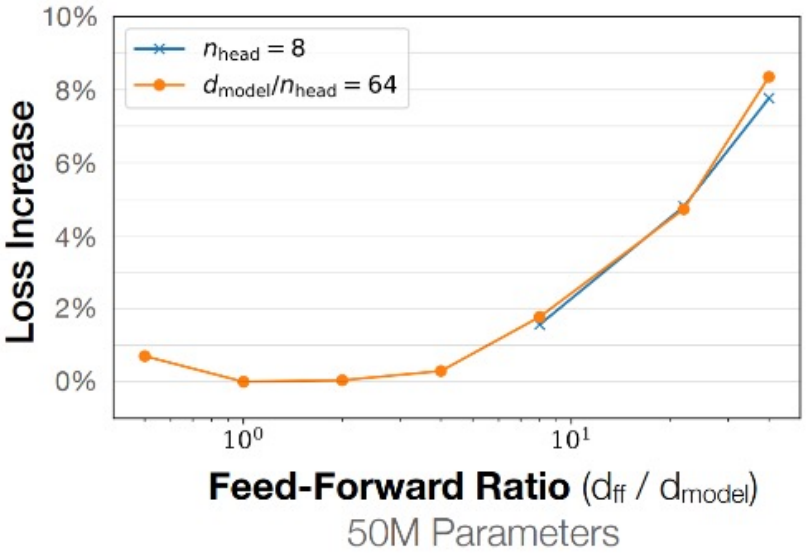
$$d_{\mathrm{ff}} = 4d$$

- This is *almost* always true. There's just a few exceptions.

[Slide credit: Tatsu Hashimoto]

Why this range of multipliers?

- Empirically, there's a basin between 1-10 where this hyperparameter is near-optimal.



[Slide credit: Tatsu Hashimoto]

Scaling Laws for Neural Language Models, 2020

Exception #1 — GLU/Gated variants

Bonus

- Remember that GLU variants scale down by 2/3 rd. This means most GLU variants have $d_{\text{ff}} = \frac{8}{3} \times d$. This is mostly what happens. Some notable such examples:

Model	d_{ff}/d_{model}
PaLM	4
Mistral 7B	3.5
LLaMA-2 70B	3.5
LLaMA 70B	2.68
Qwen 14B	2.67
DeepSeek 67B	2.68
Yi 34B	2.85
T5 v1.1	2.5

- Models are roughly in this range, though PaLM, LLaMA2 and Mistral are slightly larger

[Slide credit: Tatsu Hashimoto]

- Inlucde somewhere:
- Note: Gated models use smaller dimensions for the d_{ff} by 2/3
-

Exception #2 - T5

Bonus

- As we have (and will) see, most LMs have boring, conservative hyperparameters.
- One exception is T5 [Raffel et al 2020] which has some very bold settings.

- In particular, for the 11B model, they set

$$d_{\text{ff}} = 65{,}536$$
$$d = 1024$$

- For an astounding 64-times multiplier.

> for "11B" we use $d_{\text{ff}} = 65{,}536$ with 128-headed attention producing a model with about 11 billion parameters. We chose to scale up d_{ff} specifically because modern accelerators (such as the TPUs we train our models on) are most efficient for large dense matrix multiplications like those in the Transformer's feed-forward networks.

[Slide credit: Tatsu Hashimoto]

The Surprising Consensus #2: Model Dimension Ratio

- Remember:

$$\text{head}_i = \text{Attention}\left(\mathbf{x}\mathbf{W}_i^q, \mathbf{x}\mathbf{W}_i^k, \mathbf{x}\mathbf{W}_i^v\right)$$
$$\text{MultiHeadedAttention}(\mathbf{x}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\boldsymbol{W}^O$$

In practice, we use a reduced dimension for each head.

$$\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}_i^k \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}_i^v \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \boldsymbol{W}^O \in \mathbb{R}^{d \times d}$$

- The consensus: dim of head ($\frac{d}{h}$) x num-heads (h) = model-dim (d)
- This doesn't have to be true: we can have head-dimensions > model-dim / num-heads. The matrix ($\boldsymbol{W}^O$) can take care of projection to model-dim.
  - But most models do follow this guideline

[Slide credit: Tatsu Hashimoto]

Heads vs model dim

- Some examples of this hyperparameter:

num-heads x head-dim / model-dim

	h	d/h	d	
	Num heads	**Head dim**	**Model dim**	**Ratio**
GPT3	96	128	12288	1
T5	128	128	1024	16
T5 v1.1	64	64	4096	1
LaMDA	128	128	8192	2
PaLM	48	258	18432	1.48
LLaMA2	64	128	8192	1

- Most models have ratios around 1 — notable exceptions by some google models.

[Slide credit: Tatsu Hashimoto]

Bonus

Aspect radios

- Should my model be deep or wide? How deep and how wide?
- Most models are surprisingly consistent on this one too!

Sweet spot?

Model	d_{model}/n_{layer}
BLOOM	205
T5 v1.1	171
PaLM (540B)	156
GPT3/OPT/Mistral/Qwen	128
LLaMA / LLaMA2 / Chinchila	102
T5 (11B)	43
GPT2	33

- Note here w lth.

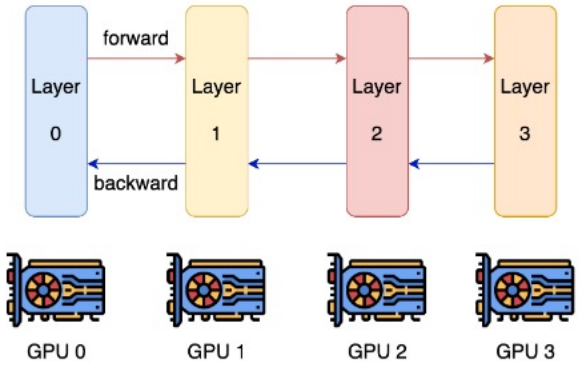
[Slide credit: Tatsu Hashimoto]

Considerations about aspect ratio

- Extremely deep models are harder to parallelize

> **The Limits of Depth vs Width** We note an obvious limitation with our advice. Scaling depth has an obvious limiter, i.e., they are non-parallelizable across different machines or devices and every computation has to always wait for the previous layer. This is unlike width, which can be easily parallelizable over thousands or hundreds of thousands of devices. Within the limitation of scaling

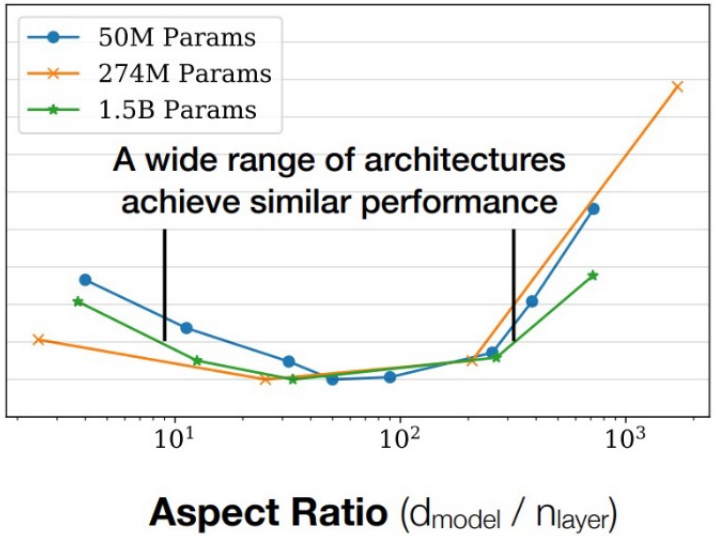
[Tay et al 2021]



[Slide credit: Tatsu Hashimoto]

Evidence on aspect ratio scaling

Wide range of ‘good’ values (100-200)



[Kaplan et al 2020]

[Slide credit: Tatsu Hashimoto]

Scaling Laws for Neural Language Models, 2020

Recap of architecture hyperparams

- Feedforward dimension / model dimension
  - Factor-of-4 rule of thumb (8/3 for GLUs) is standard (with some evidence)
- Head dim
  - Head dim*Num head = D model is standard – but not much validation
- Aspect ratio
  - Wide range of ‘good’ values (100-200). Systems concerns dictate the value.

[Slide credit: Tatsu Hashimoto]

Summary of LLM architectures

- There are many architectural variations.
- Major differences? Position embeddings, activations, tokenization
- This is an evolving field; a lot of empirical analysis is going into identifying best practices.

Name	Has pa...	Link	Year	Tokenizer type	Vocab count	Norm	Parallel Layer	Pre-norm	Position embedding	Activations	MoE	MLP factor	num_layers	model_dim
Original transformer	Yes	arxiv.org/abs....03762	2017	BPE	37000	LayerNorm	Serial	☐	Sine	ReLU	☐	4	6	
GPT	Yes	cdn.openai.com/res...er.pdf	2018	BPE	40257	LayerNorm	Serial	☐	Absolute	GeLU	☐	4	12	
GPT2	Yes	cdn.openai.com/bet...rs.pdf	2019	BPE	50257	LayerNorm	Serial	☑	Sine	GeLU	☐	4	48	
T5 (11B)	Yes	arxiv.org/abs....10683	2019	SentencePiece	32128	RMSNorm	Serial	☑	Relative	ReLU	☐	64	24	
GPT3 (175B)	Yes	arxiv.org/abs....14165	2020	BPE	50257	LayerNorm	Serial	☑	Sine	GeLU	☐	4	96	
mT5	Yes	arxiv.org/abs....11934	2020	SentencePiece	250000	RMSNorm	Serial	☑	Relative	GeGLU	☐	2.5	24	
T5 (XXL 11B) v1.1	Kind of	github.com/goo...d#t511	2020	SentencePiece	32128	RMSNorm	Serial	☑	Relative	GeGLU	☐	2.5	24	
Gopher (280B)	Yes	arxiv.org/abs....11446	2021	SentencePiece	32000	RMSNorm	Serial	☑	Relative	ReLU	☐	4	80	
Anthropic LM (not claude)	Yes	arxiv.org/abs....00861	2021	BPE	65536			☑			☐	4	64	
LaMDA	Yes	arxiv.org/abs....08239	2021	BPE	32000			☑	Relative	GeGLU	☐	8	64	
GPTJ	Kind of	huggingface.co/Ele...t-j-6b	2021	BPE	50257	LayerNorm	Parallel	☑	RoPE	GeLU	☐		28	
Chinchilla	Yes	arxiv.org/abs....15556	2022	SentencePiece	32000	RMSNorm	Serial	☑	Relative	ReLU	☐	4	80	
PaLM (540B)	Yes	arxiv.org/abs....02311	2022	SentencePiece	256000	RMSNorm	Parallel	☑	RoPE	SwiGLU	☐	4	118	
OPT (175B)	Yes	arxiv.org/abs....01068	2022	BPE	50272	LayerNorm	Serial	☑	Absolute	ReLU	☐	4	96	
BLOOM (175B)	Yes	arxiv.org/abs....05100	2022	BPE	250680	LayerNorm	Serial	☑	AliBi	GeLU	☐	4	70	
GPT-NeoX	Yes	arxiv.org/pdf...45.pdf	2022	BPE	50257	LayerNorm	Parallel	☑	RoPE	GeLU	☐	4	44	
GPT4 OPEN	Ad	arxiv.org/abs....08774	2023	BPE	100000			☐			☐			
LLaMA (65B)	Yes	arxiv.org/abs....13971	2023	BPE	32000	RMSNorm	Serial	☑	RoPE	SwiGLU	☐	2.6875	80	
LLaMA2 (70B)	Yes	arxiv.org/abs....09288	2023	BPE	32000	RMSNorm	Serial	☑	RoPE	SwiGLU	☐	3.5	80	
Mistral (7B)	Yes	arxiv.org/abs....06825	2023	BPE	32000	RMSNorm	Serial	☑	RoPE	SwiGLU	☐	3.5	32	

[Picture credit: Tatsu Hashimoto]

Another View of Architectural Variations

Name	Year	Norm	Parallel Layer	Pre-norm	Position embedding	Activations
Original transformer	2017	LayerNorm	Serial	☐	Sine	ReLU
GPT	2018	LayerNorm	Serial	☐	Absolute	GeLU
T5 (11B)	2019	RMSNorm	Serial	☑	Relative	ReLU
GPT2	2019	LayerNorm	Serial	☑	Sine	GeLU
T5 (XXL 11B) v1.1	2020	RMSNorm	Serial	☑	Relative	GeGLU
mT5	2020	RMSNorm	Serial	☑	Relative	GeGLU
GPT3 (175B)	2020	LayerNorm	Serial	☑	Sine	GeLU
GPTJ	2021	LayerNorm	Parallel	☑	RoPE	GeLU
LaMDA	2021			☑	Relative	GeGLU
Gopher (280B)	2021	RMSNorm	Serial	☑	Relative	ReLU
GPT-NeoX	2022	LayerNorm	Parallel	☑	RoPE	GeLU
BLOOM (175B)	2022	LayerNorm	Serial	☑	AliBi	GeLU
OPT (175B)	2022	LayerNorm	Serial	☑	Absolute	ReLU
PaLM (540B)	2022	RMSNorm	Parallel	☑	RoPE	SwiGLU
Chinchilla	2022	RMSNorm	Serial	☑	Relative	ReLU
Mistral (7B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
LLaMA2 (70B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
LLaMA (65B)	2023	RMSNorm	Serial	☑	RoPE	SwiGLU
Qwen (14B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
DeepSeek (67B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU
Yi (34B)	2024	RMSNorm	Serial	☑	RoPE	SwiGLU

Low consensus (except pre-norm)

Most try to follow previous successful choices.

[Slide credit: Tatsu Hashimoto]

Pre-training data

The pre-training data size and sources

- They vary quite a bit!
- They used to be in billions of tokens; now they're north of trillions.

Model Name	Release	Pre-training data #Tokens	Training Dataset
BERT	2018	3.3B	BooksCorpus (800M), English Wikipedia (2.5B)
GPT-1	2018	13B	BooksCorpus
GPT-2	2019	40B	WebText: scraping outbound links from Reddit post with >= 3 karma
T5	2019	34B	C4 which is the cleaned up version of CommonCrawl
GPT-3	2020	400B	Common Crawl (filtered), WebText2, Myrstry books!! (Books1, Books2), Wikipedia
Gopher	2021	1.4T	MassiveText
BLOOM	2022	350B	ROOTS corpus, a dataset comprising hundreds of sources in 46 natural and 13 programming languages (59 in total)
PaLM	2022	2.81T	Web documents, books, Wikipedia, conversations, GitHub code
LaMDA	2022	1.56T	Public dialog data and web documents
Chinchilla	2022	1.4T	MassiveText
LLaMA2	2023	2.0T	A new mix of publicly available online data
GPT-4	2023	?	?
Claude-3	2023	?	?
OLMo 2	2024	5.6T	OLMo-Mix-1124(stage1) + Dolmino-Mix-1124(stage 2)
Qwen2.5	2024	7T	
DeepSeek (V3)	2024	14.8T	GitHub's Markdown and StackExchange
LLaMA3	2024	15T	A new mix of publicly available online data

Where do we begin to collect data?

- Where do I find a very large dataset?
  - Crawling web is non-trivial (unless you're OpenAI or Google with ton of resources).
  - But if you have to do it, be aware that websites have their own permissions regarding which parts of their content, if any, can be crawled. (next slide)
  - The alternative is to look for websites that have done the crawling for you.

CommonCrawl



- A non-profit organization that release a new crawl of the internet **every month**.
  - So far, there have been ~100 crawls from 2008-2024.
  - In 2016, a crawl took 10-12 days on 100 machines. They used Apache Nutch.
  - This is **not** a complete of the internet. Crawls have some overlap but try to diversify.
    - Common Crawl follows links from previously crawled pages.
  - Also note, it respects robots.txt
- CC is a common sources of pre-training data.
  - **WARC**: The raw HTTP responses, including full web pages.
  - **WAT**: The metadata summary from WARC files.
  - **WET**: The extracted plaintext from WARC files, stripping out HTML and other non-textual content.

Data Type	File List	#Files	Total Size Compressed (TiB)
Segments	segment.paths.gz	100	
WARC	warc.paths.gz	90000	76.08
WAT	wat.paths.gz	90000	17.68
WET	wet.paths.gz	90000	7.00
Robots.txt files	robotstxt.paths.gz	90000	0.15
Non-200 responses	non200responses.paths.gz	90000	2.59
URL index files	cc-index.paths.gz	302	0.19
Columnar URL index files	cc-index-table.paths.gz	900	0.22

https://data.commoncrawl.org/crawl-data/CC-MAIN-2024-30/index.html

CC is messy. Is that a concern?

Besides quantity, the choice of dataset is also critical



Garbage in Garbage OUT





[Slide credit: Samet Samik]

C4: A cleaned up pre-training dataset

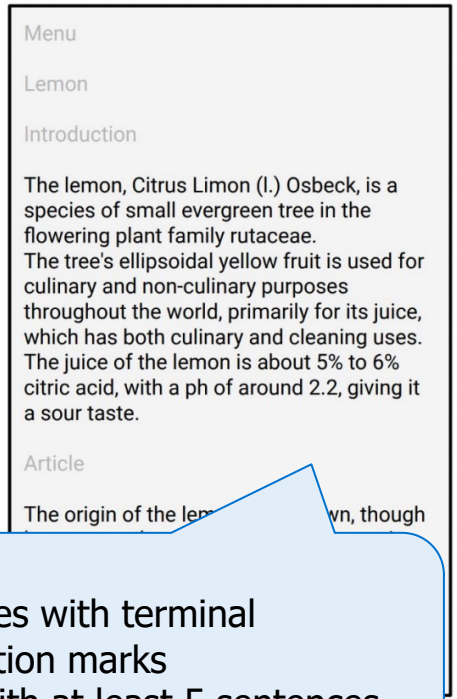
- C4: Colossal Clean Crawled Corpus
  - The course is CommonCrawl.
  - English language only
  - 750GB after ton of filtering

Data set	Size
★ C4	745GB
C4, unfiltered	6.1TB

- Notice that the unfiltered data is quite large.
  - Common Crawl is mostly not useful natural language

C4: The D

Remove any:
- References to Javascript
- Pages with "{" (no code), "Lorem ipsum" text (dummy text), "terms of use", etc.
- Pages with "bad words".



Menu

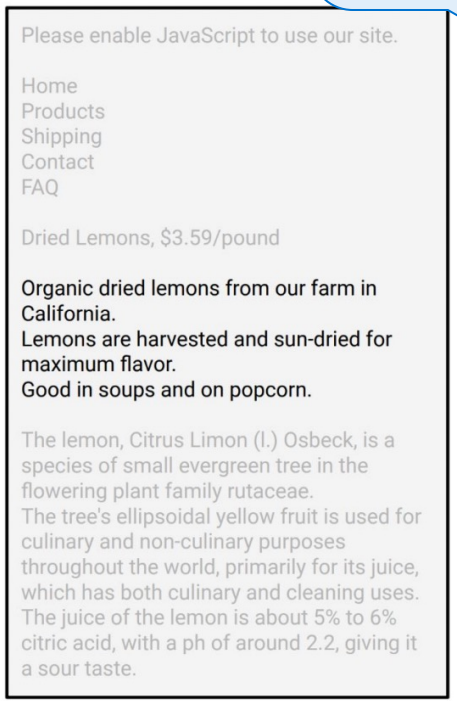
Lemon

Introduction

The lemon, Citrus Limon (l.) Osbeck, is a species of small evergreen tree in the flowering plant family rutaceae.
The tree's ellipsoidal yellow fruit is used for culinary and non-culinary purposes throughout the world, primarily for its juice, which has both culinary and cleaning uses. The juice of the lemon is about 5% to 6% citric acid, with a ph of around 2.2, giving it a sour taste.

Article

The origin of the lem wn, though



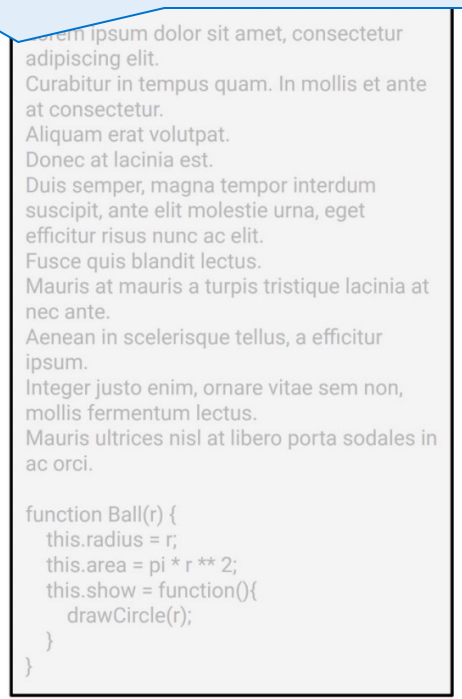
Please enable JavaScript to use our site.

Home
Products
Shipping
Contact
FAQ

Dried Lemons, $3.59/pound

Organic dried lemons from our farm in California.
Lemons are harvested and sun-dried for maximum flavor.
Good in soups and on popcorn.

The lemon, Citrus Limon (l.) Osbeck, is a species of small evergreen tree in the flowering plant family rutaceae.
The tree's ellipsoidal yellow fruit is used for culinary and non-culinary purposes throughout the world, primarily for its juice, which has both culinary and cleaning uses. The juice of the lemon is about 5% to 6% citric acid, with a ph of around 2.2, giving it a sour taste.



orem ipsum dolor sit amet, consectetur adipiscing elit.
Curabitur in tempus quam. In mollis et ante at consectetur.
Aliquam erat volutpat.
Donec at lacinia est.
Duis semper, magna tempor interdum suscipit, ante elit molestie urna, eget efficitur risus nunc ac elit.
Fusce quis blandit lectus.
Mauris at mauris a turpis tristique lacinia at nec ante.
Aenean in scelerisque tellus, a efficitur ipsum.
Integer justo enim, ornare vitae sem non, mollis fermentum lectus.
Mauris ultrices nisl at libero porta sodales in ac orci.

```
function Ball(r) {
  this.radius = r;
  this.area = pi * r ** 2;
  this.show = function(){
    drawCircle(r);
  }
}
```

Retain:
- Sentences with terminal punctuation marks
- Pages with at least 5 sentences, sentences with at least 3 words

Slide adapted from Colin Raffel

Pre-training Data: Experiment

- Takeaway:
  - Clean and compact data is better than large, but noisy data.
  - Pre-training on in-domain data helps.

Data set	Size	GLUE	CNNDM	SQuAD	SGLUE	EnDe	EnFr	EnRo
★ C4	745GB	83.28	**19.24**	80.88	71.36	**26.98**	**39.82**	**27.65**
C4, unfiltered	6.1TB	81.46	19.14	78.78	68.04	26.55	39.34	27.21

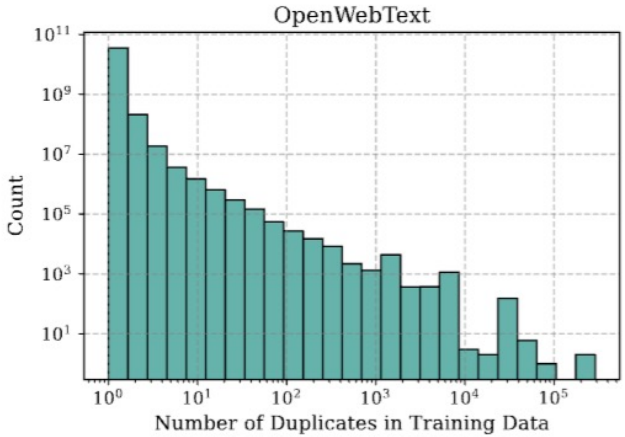
Exploring the limits of transfer learning with text-to-text transfer transformers, 2020

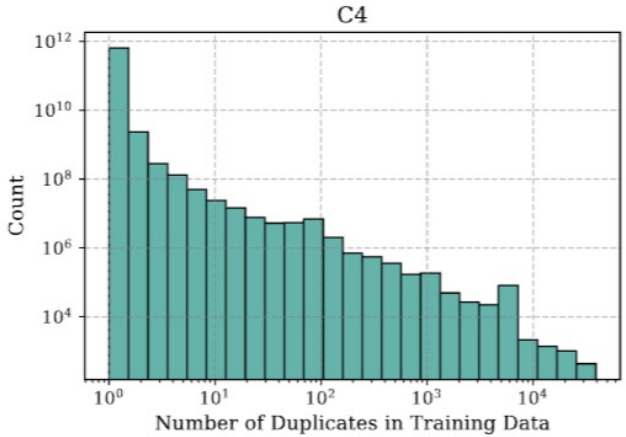
Does it matter that my data has ton of repetitions?

Pre-training Data Duplicates

- There is a non-negligible number of duplicates in any pre-training data.

	% train examples with dup in train	% train examples with dup in valid	% valid with dup in train
C4	3.04%	1.59%	4.60%
RealNews	13.63%	1.25%	14.35%
LM1B	4.86%	0.07%	4.92%
Wiki40B	0.39%	0.26%	0.72%





Deduplicating Training Data Mitigates Privacy Risks in Language Models, 2022

Pre-training Data Duplicates

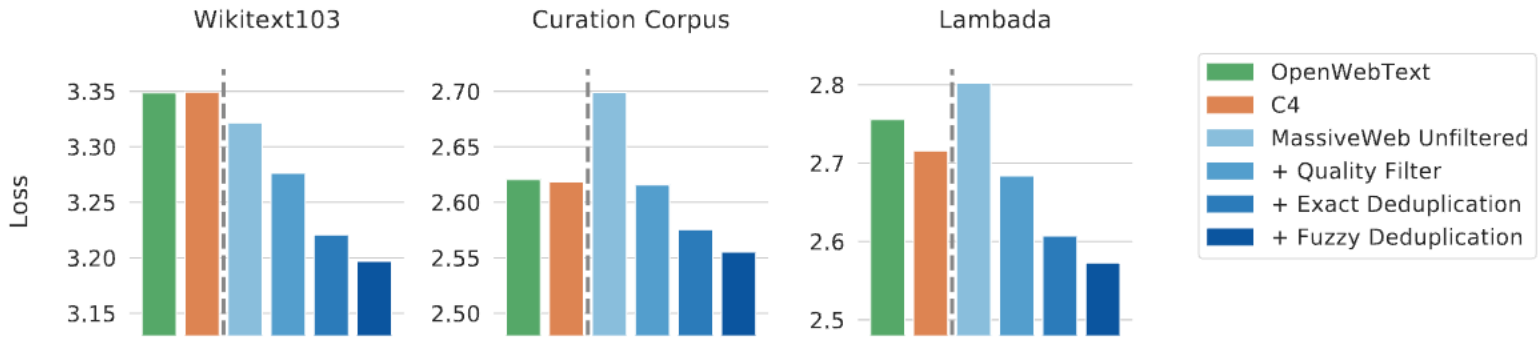
- There is a non-negligible number of duplicates in any pre-training data.
- Maybe we should not spend our training budget re-learning things we have already seen.

Dataset	Example	Near-Duplicate Example
Wiki-40B	\n_START_ARTICLE_\nHum Award for Most Impactful Character \n_START_SECTION_\nWinners and nominees\n_START_PARAGRAPH_\nIn the list below, winners are listed first in the colored row, followed by the other nominees. [...]	\n_START_ARTICLE_\nHum Award for Best Actor in a Negative Role \n_START_SECTION_\nWinners and nominees\n_START_PARAGRAPH_\nIn the list below, winners are listed first in the colored row, followed by the other nominees. [...]
LM1B	I left for California in 1979 and tracked Cleveland 's changes on trips back to visit my sisters .	I left for California in 1979 , and tracked Cleveland 's changes on trips back to visit my sisters .
C4	Affordable and convenient holiday flights take off from your departure country, "Canada". From May 2019 to October 2019, Condor flights to your dream destination will be roughly 6 a week! Book your Halifax (YHZ) - Basel (BSL) flight now, and look forward to your "Switzerland" destination!	Affordable and convenient holiday flights take off from your departure country, "USA". From April 2019 to October 2019, Condor flights to your dream destination will be roughly 7 a week! Book your Maui Kahului (OGG) - Dubrovnik (DBV) flight now, and look forward to your "Croatia" destination!

Deduplicating Training Data Makes Language Models Better, 2020

Deduplicating Data Improves LMs

- Another evidence from Gopher paper: Performance of 1.4B parameter models (lower is better) trained on OpenWebText, C4, and versions of MassiveWeb with progressively more pre-processing stages added.
- Applying a quality filter and de-duplication stages significantly improves quality.



Scaling Language Models: Methods, Analysis & Insights from Training Gopher, 2022

How can I do my own deduplication?

How do you scale data deduplication?

- Pre-training is huge. Naively deduplicating the data is going to take forever!!
- How do you deduplicate it? Here are a few options:
  - Naively hashing each document (a good baseline)
  - SuffixArray
  - MinHash
  - BloomFilters
  - Embedding-based dedup

Comparison between dedup algorithms

- Single methods: BF better than any other method standalone.
- Combination: The competitive approaches are last row (exact -> MH -> SA) and BF-only. The former leads to more compact data.

Exact Dedup	MinHash	Suffix Array	Bloom Filter	Tokens	Removal Rate	CORE	Δ from Baseline
✗	✗	✗	✗	76B	00%	40.1	+0.0
✓	✗	✗	✗	66B	13%	41.0	+0.9
✗	✓	✗	✗	62B	18%	40.9	+0.8
✗	✗	✓	✗	51B	33%	41.4	+1.3
✗	✗	✗	✓	**56B**	**26%**	**41.7**	**+1.6**
✓	✓	✗	✗	58B	24%	40.2	+0.1
✓	✗	✓	✗	49B	36%	41.3	+1.3
✗	✓	✓	✗	48B	37%	41.2	+1.2
✓	✓	✓	✗	**45B**	**41%**	**41.7**	**+1.6**

Individual technique

Combined techniques

Li et al. DataComp-LM: In search of the next generation of training sets for language models, 2024

Deduplication: Recap

Bonus

- Does it matter that my data has ton of repetitions? Yes, one should do careful dedup.
- How can I do my own deduplication?
  - Scaling it up requires advanced data structures.
  - So far, there is no clear winner between these algorithms. A “kitchen sink” approach that mixes dedup algorithms is generally best, but it’s an empirical exercise.
  - BF is generally preferred since it’s cheaper/faster.

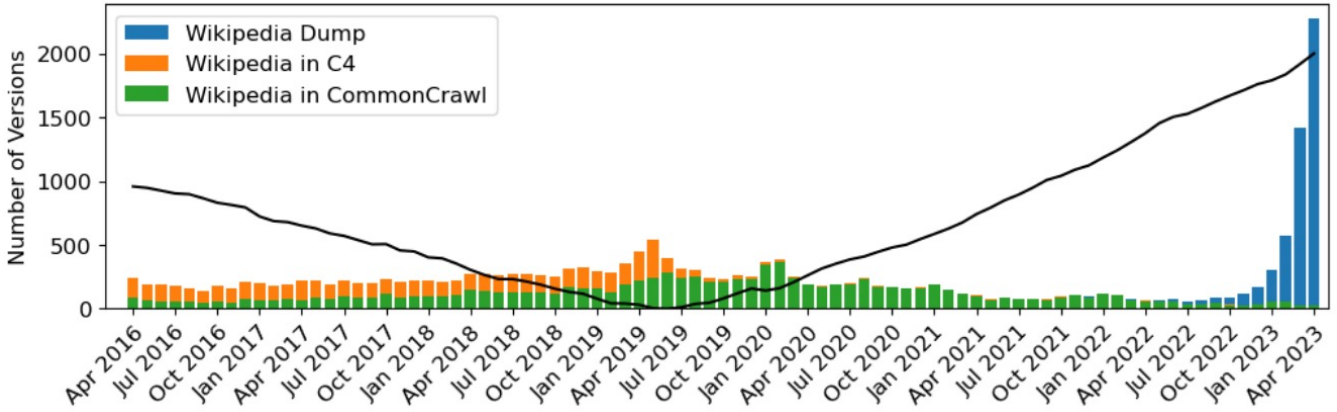
Bonus

Should I worry about old data in my pre-training?

Prevalence of stale data: RedPejamas

Bonus

- Breakdown of old versions of Wikipedia in RedPejamas
  - It is based on dumps from C4, CC and a recent Wikipedia dump.
- The bars blow show the breakdown of older versions of Wikipedia in RedPajamas.
  - There is a ton of old Wikipedia versions in RedPejamas! 😨
- The solid trend is the perplexity of a pre-trained model on temporal instances of Wikipedia.
  - The significant stale training data in has skewed PPL toward older versions of Wikpedia.



Dated Data: Tracing Knowledge Cutoffs in Large Language Models, 2024

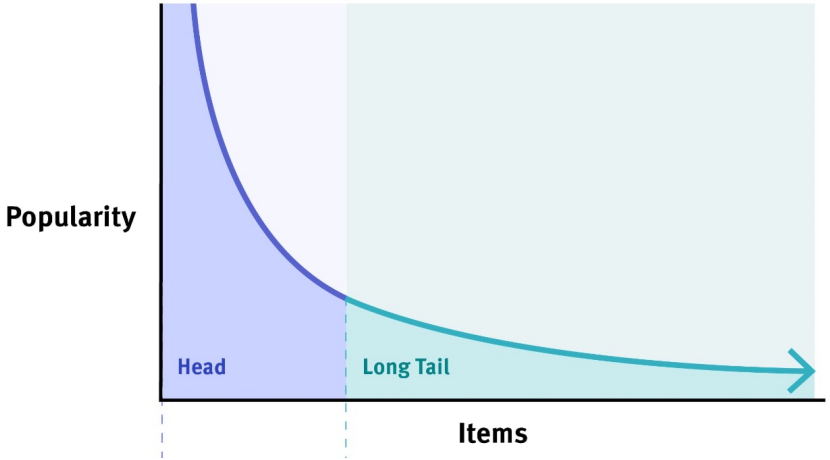
Bonus

Should I worry about skew of the data mixtures in my pre-training?

Bonus

Data mixtures (and the long tail)

- Your dataset mixture will determine the versatility of the resulting model.
- Data in the world is always skewed. For example,
  - English has a lot more language than other domains.
  - Reddit is a lot larger than science papers.



Upsample or Upweight? Balanced Training on Heavily Imbalanced Datasets, 2024
DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining, 2023

Data mixtures (and the long tail)

Bonus

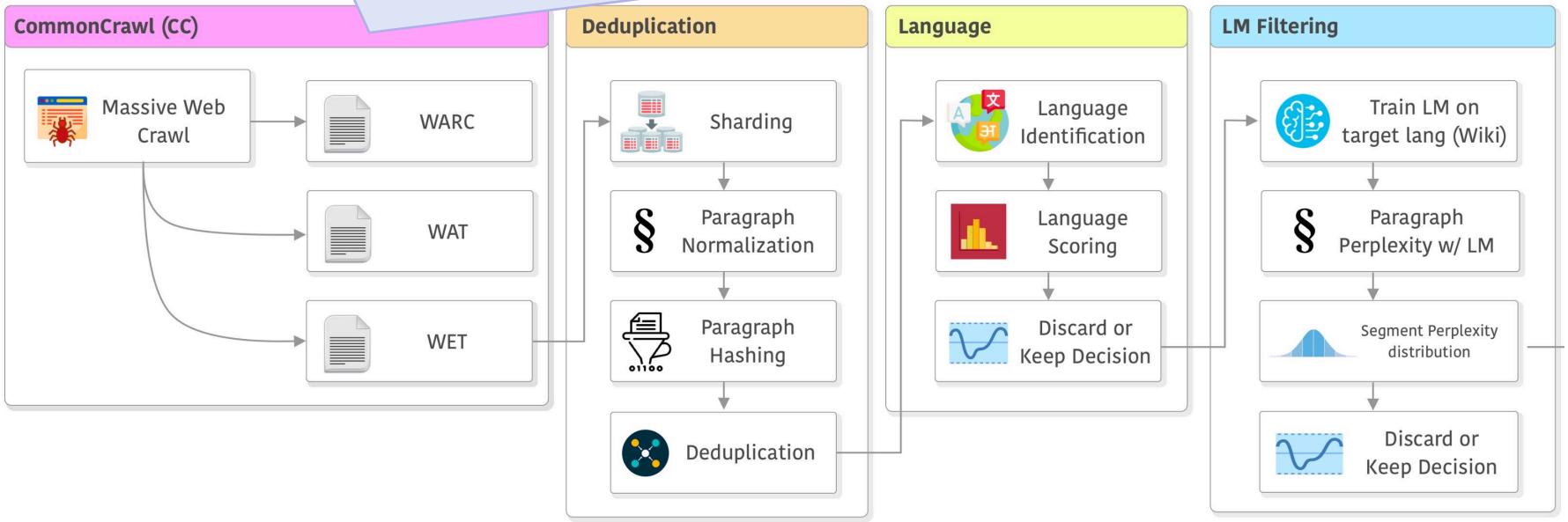
- Your dataset mixture will determine the versatility of the resulting model.
- Data in the world is always skewed. For example,
  - English has a lot more language than other domains.
  - Reddit is a lot larger than science papers.
- A uniform ”weight” of data during pre-training is not good since overrepresented domains would dominate (e.g., your model would be a better at English than Azeri).
- Overamplifying underrepresented domains also runs risk of overfitting.
- So, there is a lot of research on finding **a good balance**.

Upsample or Upweight? Balanced Training on Heavily Imbalanced Datasets, 2024
DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining, 2023

Few notable data pipelines

LLaMA 1's Data Pipeline

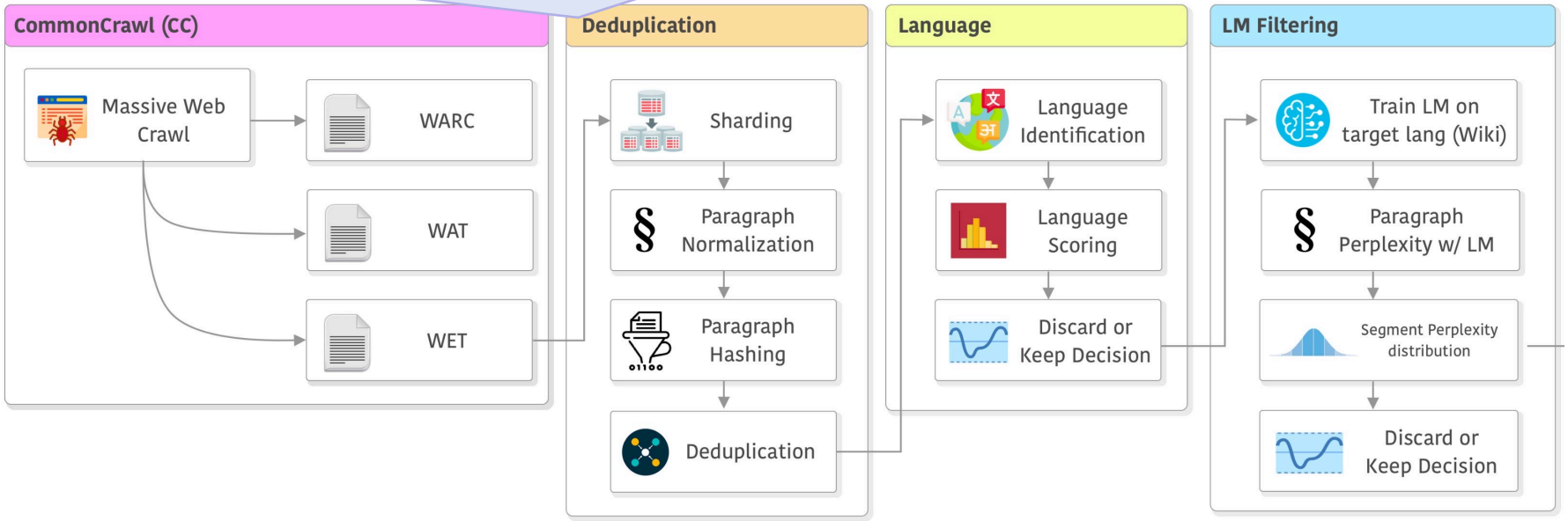
Starts with the massive crawled data by CommonCrawl.
The WET format that contains textual information.
WARC is raw, WAT is metadata, WET is text+some metadata.



https://twitter.com/tarantulae/status/1650170087708454913?t=ncWWY0FI0tYC_dYLs76r5g

LLaMA 1's Data Pipeline

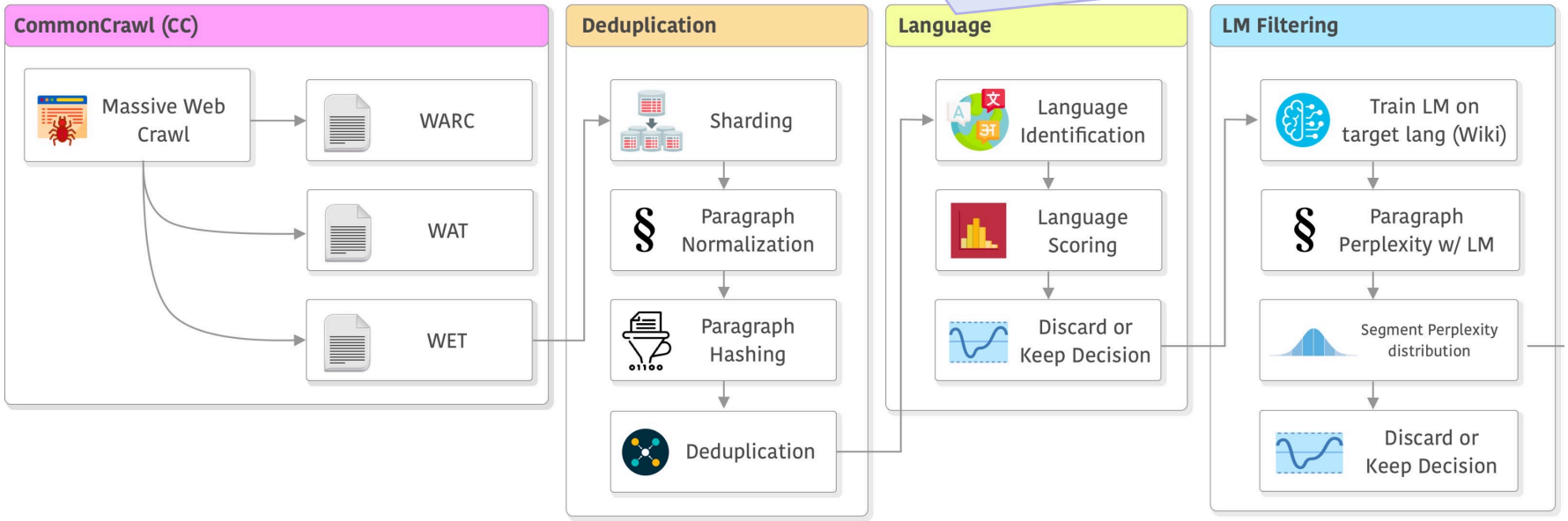
Shard WET content into shards of 5GB each (one CC snapshot can have 30TB). Then you normalize paragraphs (lowercasing, numbers as placeholders, etc), compute per-paragraph hashes and then duplicate them.



https://twitter.com/tarantulae/status/1650170087708454913?t=ncWWY0FI0tYC_dYLs76r5g

LLaMA 1's Data Pipeline

Perform language identification and decide whether to keep or discard languages.
The order of when you do this in the pipeline can impact the language discrimination quality.

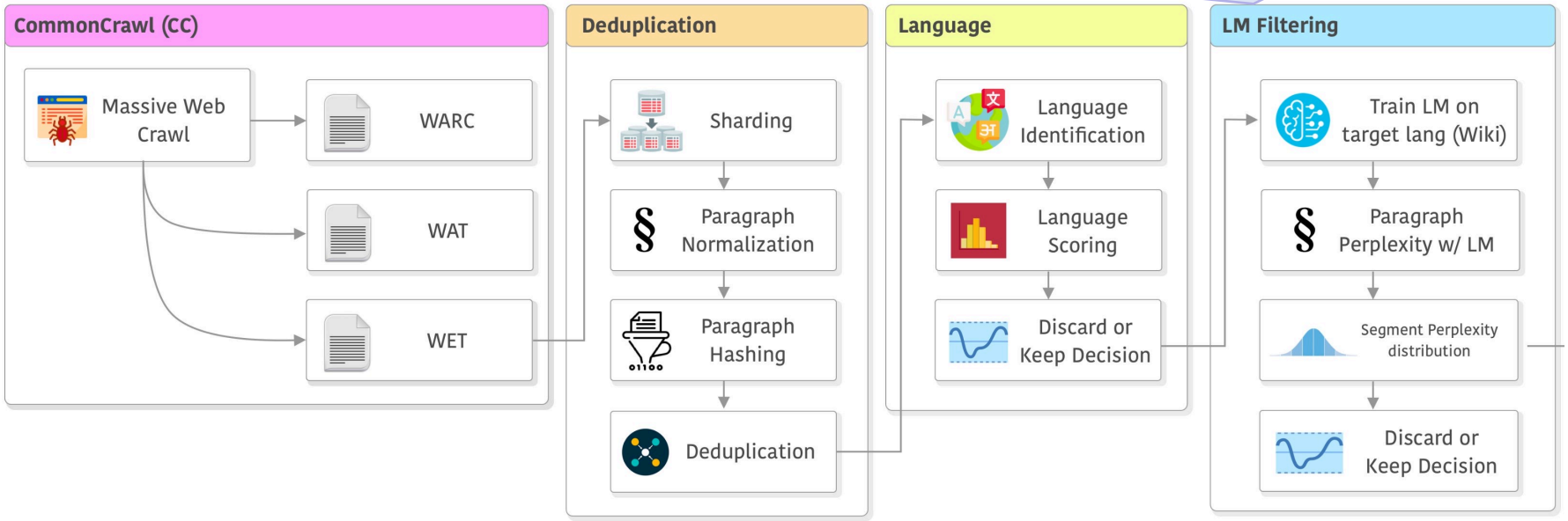


https://twitter.com/tarantulae/status/1650170087708454913?t=ncWWY0FI0tYC_dYLs76r5g

LLaMA 1's Data Pipeline

Do further quality filtering: Train a simple LM (n-gram) on target languages using Wikipedia, then compute per-paragraph perplexity on the rest of the data:

- Very high PPL: Very different than Wiki and likely low-quality → Drop
- Very low PPL: Very similar or near duplicates to Wiki → Drop



CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data, 2019

DataDecomp-LM filtering pipeline

Bonus

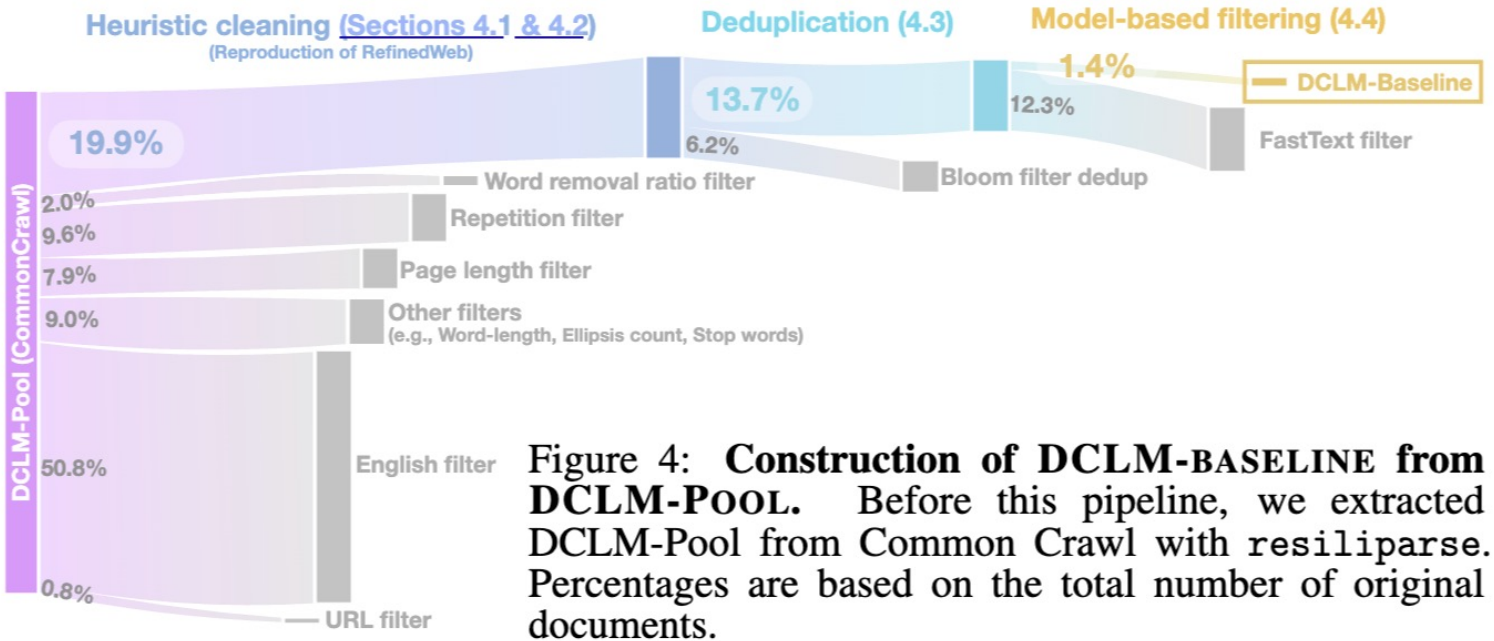


Figure 4: **Construction of DCLM-BASELINE from DCLM-POOL.** Before this pipeline, we extracted DCLM-Pool from Common Crawl with `resiliparse`. Percentages are based on the total number of original documents.

DataComp-LM: In search of the next generation of training sets for language models, 2024

Few cleaned-up pre-training datasets

Bonus

Dataset	Example models	Tokens	Source	License	Lang
C4 (Raffel et al. 2020)	T5	165B	CC	ODC-BY	English
The Pile (Gao el al. 2020)	GPT-J, Pythia	300B	22 datasets including CC, books, code, news	Varies by dataset subset	English
RedPejamas (Weber et al. 2024)	Llama 1	1.2T	CC, C4, Github, Arxiv, Books, Wikipedia, StackExchange	Varies by dataset subset	English
RefinedWeb (Penedo et al. 2023)	Falcon	600B	CC	ODC-BY 1.0	English
Dolma (Soldaini et al. 2024)	OLMo	3T	CC, C4, Gutenberg, Github, Wikipedia, Wikibooks	ImpACT MR	English
DataComp-LM (Li et al. 2024)	SmolLM2, DCLM	240T	CC	?	English

The Pile

Bonus

- Pile-CC: From Common Crawl; uses justText to extract useful text.
- PubMed Central: 5M NIH funded papers and public.
- arXiv: preprint for research papers since 1991 (uses latex).
- Gutenberg PG-19: Online books (before 2019) with copyright clearance.
- Books3 is a a collection of ~200K books. Has been subject of lawsuits.
- StackExachange: Q&A format is close to real applications.
- Github: Content is not just the code.
  - Note, GH archive has regular snapshots of Github (commits, forks, etc.)

Component	Raw Size
Pile-CC	227.12 GiB
PubMed Central	90.27 GiB
Books3†	100.96 GiB
OpenWebText2	62.77 GiB
ArXiv	56.21 GiB
Github	95.16 GiB
FreeLaw	51.15 GiB
Stack Exchange	32.20 GiB
USPTO Backgrounds	22.90 GiB
PubMed Abstracts	19.26 GiB
Gutenberg (PG-19)†	10.88 GiB
OpenSubtitles†	12.98 GiB
Wikipedia (en)†	6.38 GiB
DM Mathematics†	7.75 GiB
Ubuntu IRC	5.52 GiB
BookCorpus2	6.30 GiB
EuroParl†	4.59 GiB
HackerNews	3.90 GiB
YoutubeSubtitles	3.73 GiB
PhilPapers	2.38 GiB
NIH ExPorter	1.89 GiB
Enron Emails†	0.88 GiB
The Pile	**825.18 GiB**

The Pile: An 800GB Dataset of Diverse Text for Language Modeling, 2020

Slide inspiration: Percy Liang

Summary: preparing pre-training data

- Data does not fall from the sky. You have to work to get it!
- **Finding large data:** CommonCrawl has a ton of crawled dumps, but not the only one.
- **Cleaning data** can save tons of compute and even give you gains.
- **Repetitions** are often a waste of compute and deteriorate model quality.
- **Scaling deduplication** requires advanced data structures.
- **Old data** old data may skew your model predictions, but it depends on your application.
- **Data mixtures** are quite important, though depend on your downstream application.

The actual pre-training

How should we select the right hyperparams?

Q: What would you do?

- Zuckerberg gave you a $500M budget for training Llama-10.
- You set aside $10M for finding the best architecture at smaller scale, assuming that your ultimate model will be much larger.
- This way, you pick your parameters with rigorous experiments at small scale:
  - Optimal training params: Learning rate, warmup, weight decay, etc.
  - Architecture configs by scaling (each x50) the optimal values at small scale.

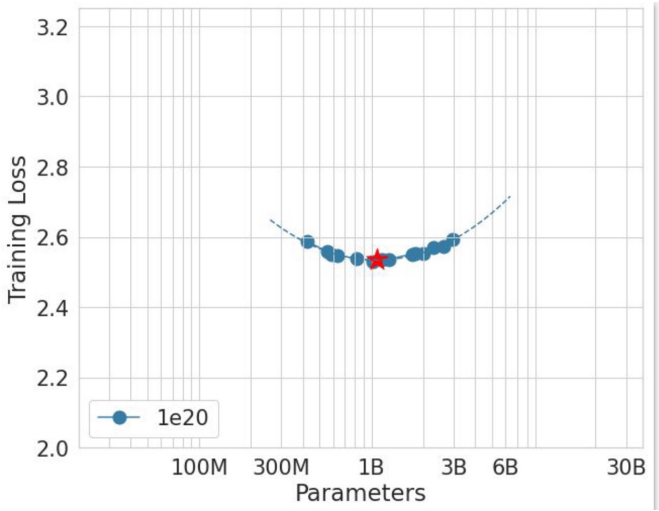
- Q: What you like (or don't) about this recipe?

- Optimal depth/width, lr, batch size, weight decay, init, and residual scaling are **not scale-invariant**.

IsoPlots: Tradeoffs at a smaller scale

- The performance of your model depends on a complex combination of many factors.
- Goal: find the best combinations, for a fixed compute.
- Approach:
  1. Fix a compute budget FLOP
  2. Train a few models and vary their size
  3. Fit a parabola and find the minimum

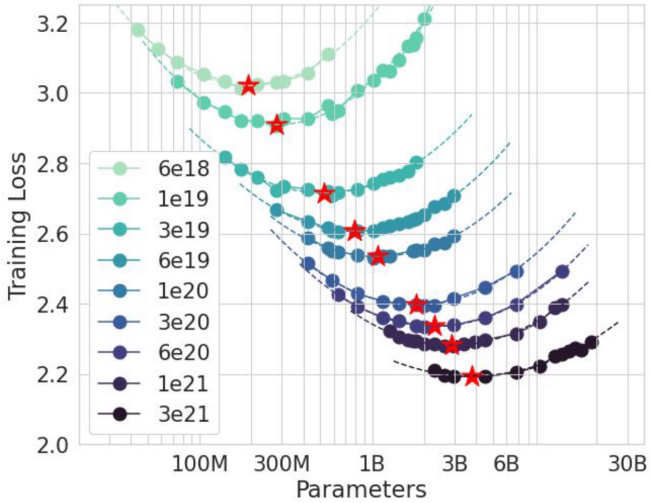
IsoPlots (Isocontours)



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  4. Repeat 1-3 for various FLOP budgets

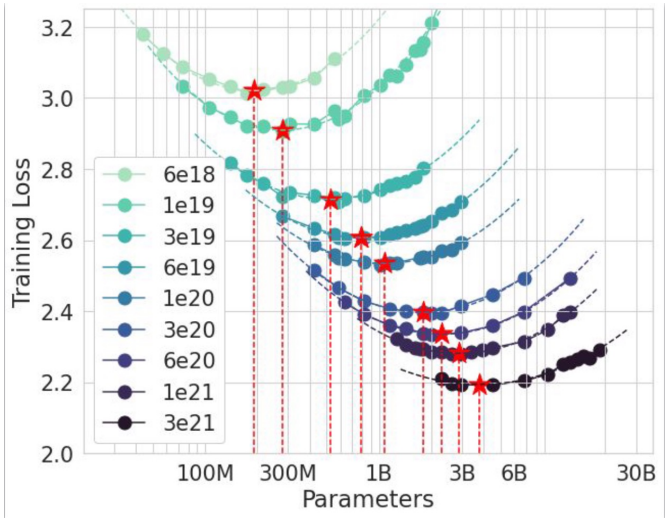
IsoPlots (Isocontours)



IsoPlots: Tradeoffs at a smaller scale

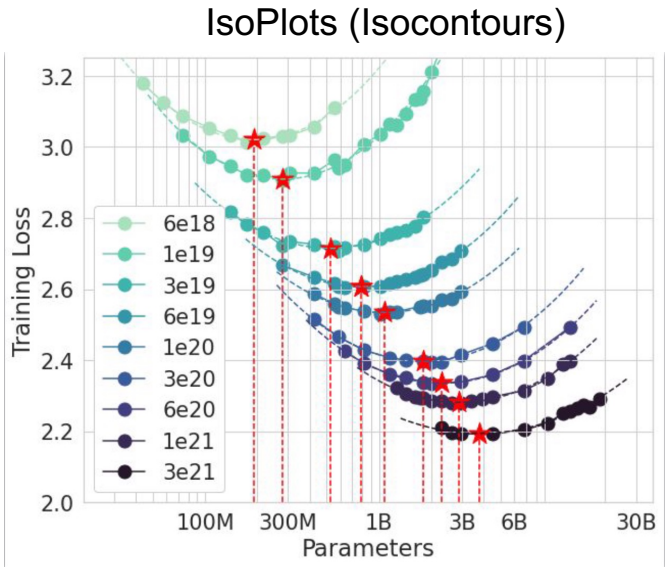
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- Goal: find the best combinations, for a fixed compute.
- Approach:
  1. Fix a compute budget FLOP
  2. Train a few models and vary their size
  3. Fit a parabola and find the minimum
  4. Repeat 1-3 for various FLOP budgets
- It's good to change various parameter (e.g., training data, size, or other hyperparams) and see how it's quality (loss) changes.

IsoPlots (Isocontours)



Predictive models for parameters

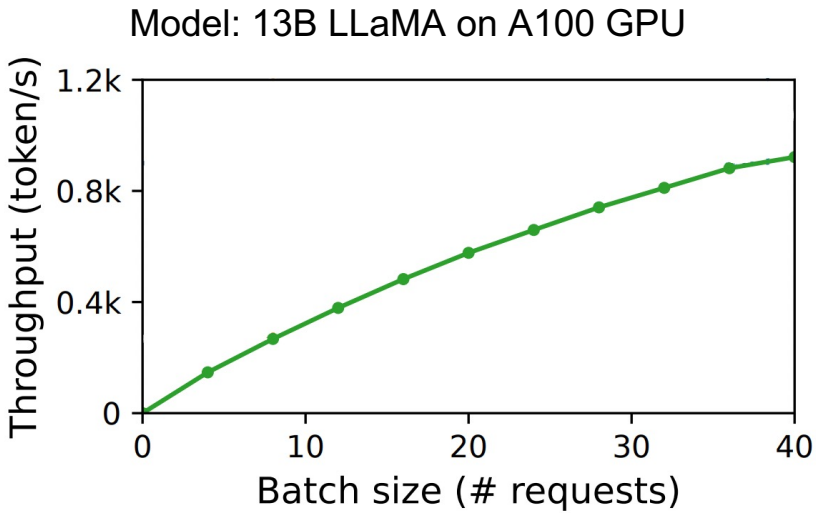
- Overall, IsoPlots show how loss depends on three axes: model size (parameters N), dataset size (tokens D), and compute budget C.
- They're a tool for reasoning about *how to spend your training budget efficiently*.
- But it also allows one to see track effective hyperparams (LR, batch size, etc.) changes with N, C, or D.
- **Lesson:**
  - Don't overtune your hyperparameters at small scales and expect to use them at large
  - Instead develop predictive metrics based on parameters:



How should I train the model?

Batching Data

- Previously we talked about the importance of batching data
- GPUs are faster at Tensor operations and hence, we want to do batch processing
- The lager batch of data, the faster they get processed.
- Alas, the speedup is often sub-linear (e.g., 2x larger batch leads to less than 2x speedup).
- If you can afford larger batch (larger GPU), it's generally worth it.



Efficient Memory Management for Large Language Model Serving with PagedAttention, 2023

Q: Agreed or not

- “Use the largest batch size that you can, at all costs” – agreed or not?

- To allocate more batches of data within your memory, you may need to truncate each instance shorter.
- The other issue is about the quality of your estimated gradients:
  - Small batch sizes will result in noisy gradients estimates.
    - Therefore, the model may not be able to converge to the optimal performances.
  - A large batch size while giving very accurate gradient estimations will tend to make less use of each training token
    - Slower convergence and potentially wasting compute.

An Empirical Model of Large-Batch Training

Batch sizes: some known statistics

LLaMA: Open and Efficient Foundation Language Models, 2023

params	dimension	n heads	n layers	learning rate	batch size	n tokens
6.7B	4096	32	32	$3.0e^{-4}$	4M	1.0T
13.0B	5120	40	40	$3.0e^{-4}$	4M	1.0T
32.5B	6656	52	60	$1.5e^{-4}$	4M	1.4T
65.2B	8192	64	80	$1.5e^{-4}$	4M	1.4T

Their seq length: 2,048 tokens

Note: tokens per batch = batch size x seq len

The Llama 3 Herd of Models, 2024

GPUs	TP	CP	PP	DP	Seq. Len.	Batch size/DP	Tokens/Batch	TFLOPs/GPU	BF16 MFU
8,192	8	1	16	64	8,192	32	16M	430	43%
16,384	8	1	16	128	8,192	16	16M	400	41%
16,384	8	16	16	8	131,072	16	16M	380	38%

Table 4 Scaling configurations and MFU for each stage of Llama 3 405B pre-training. See text and Figure 5 for descriptions of each type of parallelism.

DeepSeek-V2: A Strong, Economical, and Efficient Mixture-of-Experts Language Model, 2024

is set to 1.0. We do not employ the batch size scheduling strategy for it, and it is trained with a constant batch size of 4608 sequences. During pre-training, we set the maximum sequence

An Empirical Model of Large-Batch Training, 2018

Dropout and other regularization

- Do we need regularization during pretraining?

- Arguments against:
  - There is *a lot* of data (trillions of tokens), more than parameters.
  - SGD only does a single pass on a corpus (hard to memorize)

- This is all quite reasonable…. but what do people do in practice?

[Slide credit: Tatsu Hashimoto]

Dropout and weight decay in practice

Bonus

Model	Dropout*	Weight decay
Original transformer	0.1	0
GPT2	0.1	0.1
T5	0.1	0
GPT3	0.1	0.1
T5 v1.1	0	0
PaLM	0	(variable)
OPT	0.1	0.1
LLaMA	0	0.1
Qwen 14B	0.1	0.1

Many older models used dropout during pretraining

Newer models (except Qwen) rely only on weight decay

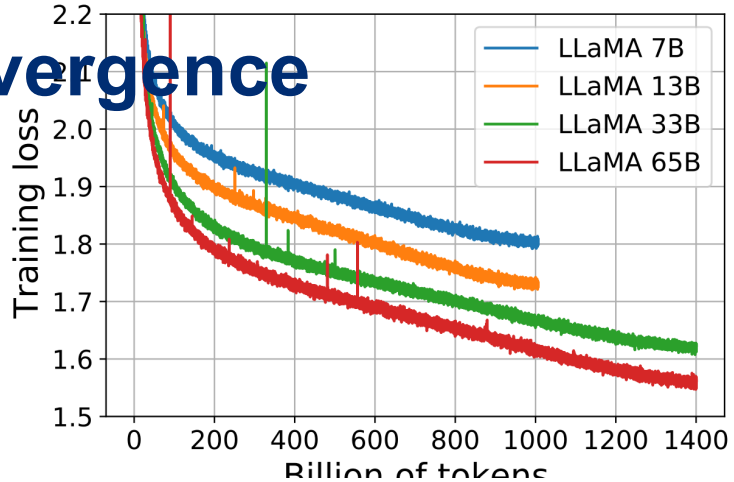
* Most of the times papers just don't discuss dropout. On open models, this closely matches not doing dropout. This may not be true of closed models.

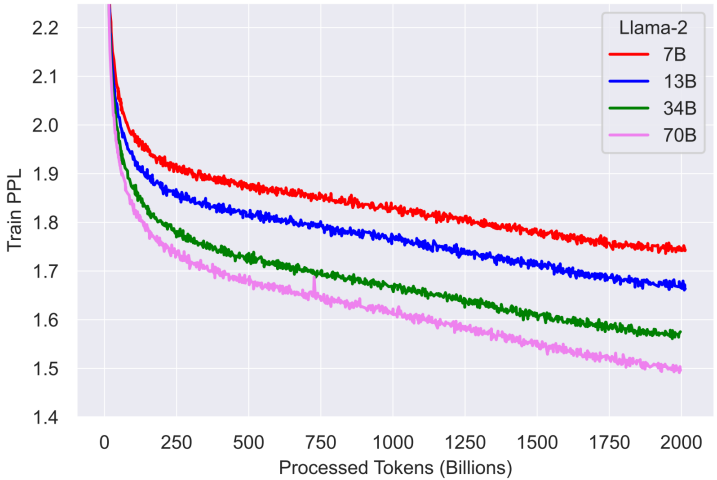
[Slide credit: Tatsu Hashimoto]

- https://arxiv.org/pdf/2509.01440

Monitoring the convergence

- In practice, your model's loss should continue to go down with more training on more data.
- So, the real bottlenecks are:
  - (1) compute
  - (2) data
- Sometimes training diverges (spikes in the loss), at which point practitioners usually restart training from an earlier checkpoint.





Monitoring the convergence with end tasks

- Some works have also monitored **end task** performance during pre-training.
- Use **likelihood** of the correct answer rather than accuracy
  - (you don't even need to consider the incorrect answers)
- Very similar to the "cloze mmlu" trend where you use the probability of the full answer instead of A, B, C, D.
  - Not discrete metrics (e.g., Accuracy)

Understanding Emergent Abilities of Language Models from the Loss Perspective

Monitoring the convergence with end tasks

- These two typically correlate, but not always.

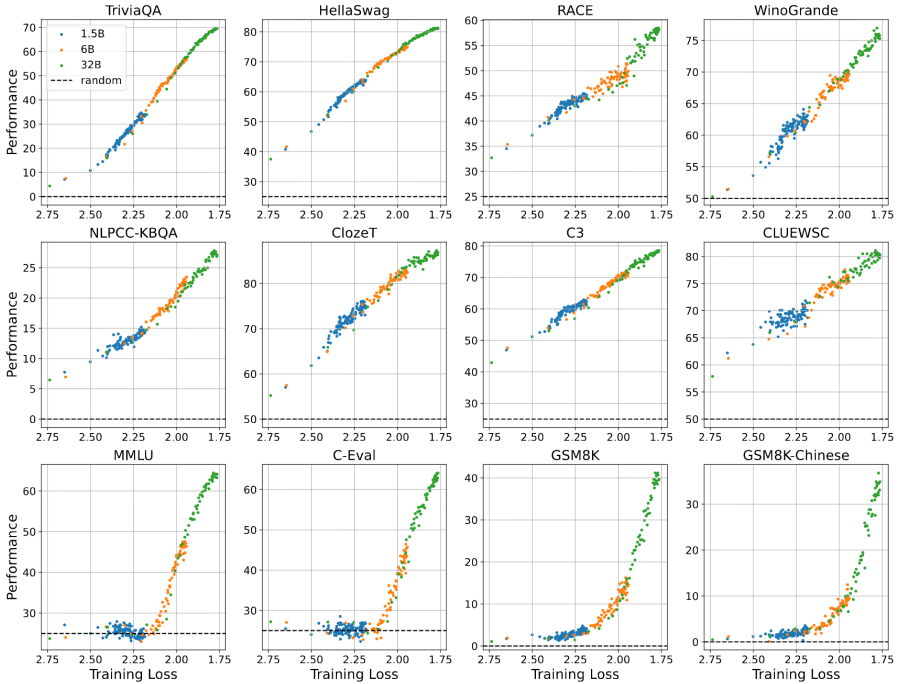


Figure 1: **The performance-vs-loss curves of 1.5B, 6B, and 32B models.** Each data point is the loss (x-axis) and performance (y-axis) of the intermediate checkpoint of one of the three models. We mark the results of random guess in black dashed lines.

[Understanding Emergent Abilities of Language Models from the Loss Perspective]

Recap of training LLMs

- **IsoPlots:** for a fixed compute, which combination of parameters give you the best bang for the buck.
- Careful batching makes your training go brrr!
- Memory usage can be tricky since there are various moving parts.
  - More on distributed training later on.
- Dropout is less common but you still ‘regularize’ LMs via large-scale training.

- Lots of cool MOE + RL stuff: https://cursor.com/resources/Composer2.pdf

- Packing docs during pre-training: Inter-document masking (also called **sample packing with attention masking**) is widely used in modern LLM pretraining and finetuning.
- https://claude.ai/share/9dcd5054-556f-4929-a04a-717f04b25e9a

Karpathy Zero to Hero

https://www.youtube.com/playlist?list=PLAqhIrjkxbuWI23v9cThsA9GvCAUhRvKZ

Nate’s slides

- https://docs.google.com/presentation/d/1oWoIrRXzLONu_oSXRG3lJdRRwIO3kkKmg37mLQtSfi4/edit?usp=sharing
- Some of these will go to the efficiency deck.
- Some others (like small models) will go here.